

**LAPORAN PRAKTIKUM
ADMINISTRASI BASIS DATA**

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**PROGRAM STUDI PENDIDIKAN TEKNOLOGI INFORMASI
FAKULTAS SAINS, TEKNIK, DAN TERAPAN
UNIVERSITAS PENDIDIKAN MANDALIKA
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MODUL 1

STUDI KASUS, ERD, DAN RELATIONAL MAPPING

Studi Kasus :

Studi Kasus: Sistem Informasi Rumah Sakit "Medika Digital"

Rumah Sakit Medika Digital ingin mendigitalisasi seluruh proses operasionalnya. Di rumah sakit ini, terdapat berbagai **Dokter** yang bertugas. Setiap dokter memiliki data berupa ID Dokter, Nama, Spesialisasi, dan Nomor Telepon. Setiap dokter wajib bernaung di bawah satu **Poliklinik** (misalnya: Poli Jantung, Poli Gigi), namun satu poliklinik bisa menaungi banyak dokter. Poliklinik sendiri diidentifikasi dengan Kode Poli dan Nama Poli.

Pasien yang datang harus mendaftarkan diri dengan memberikan data NIK, Nama, Alamat, dan Tanggal Lahir. Seorang pasien bisa melakukan banyak **Pendaftaran** untuk berkonsultasi. Setiap pendaftaran dicatat berdasarkan nomor antrean dan tanggalnya. Dalam satu pendaftaran, seorang pasien hanya berkonsultasi dengan satu dokter, dan seorang dokter bisa melayani banyak pendaftaran pasien dari waktu ke waktu.

Setelah konsultasi, dokter akan mengeluarkan **Rekam Medis**. Satu rekam medis hanya merujuk pada satu pertemuan (pendaftaran) spesifik. Di dalam rekam medis ini, dokter bisa meresepkan banyak **Obat**, dan satu jenis obat bisa muncul di banyak rekam medis pasien yang berbeda. Rumah sakit perlu mencatat jumlah (kuantitas) obat yang diberikan pada setiap rekam medis tersebut. Setiap obat memiliki Kode Obat, Nama Obat, dan Harga.

Selain itu, terdapat aturan mengenai kategori staf:

- Setiap **Staf** di rumah sakit (termasuk dokter dan perawat) harus terdaftar dalam database dengan ID Staf dan Jabatan.
- Rumah sakit menegaskan bahwa **semua** staf yang terdaftar harus masuk ke dalam salah satu kategori: apakah dia **Tenaga Medis** (dokter/perawat) atau **Tenaga Administrasi**. Tidak ada staf yang boleh tidak memiliki kategori, dan tidak mungkin seseorang menjadi tenaga medis sekaligus tenaga administrasi.
- Khusus untuk Tenaga Medis, harus mencatat Nomor STR (Surat Tanda Registrasi), sedangkan Tenaga Administrasi mencatat Locket Penugasan.

Terakhir, pasien bisa memilih untuk menginap di **Kamar**. Kamar memiliki Nomor Kamar dan Tipe (VIP/Ekonomi). Satu kamar bisa ditempati oleh banyak pasien secara bergantian (pada waktu berbeda), namun pada satu waktu rawat inap, seorang pasien hanya menempati satu kamar. Tidak semua pasien yang datang harus menginap di kamar (hanya yang memerlukan rawat inap saja).

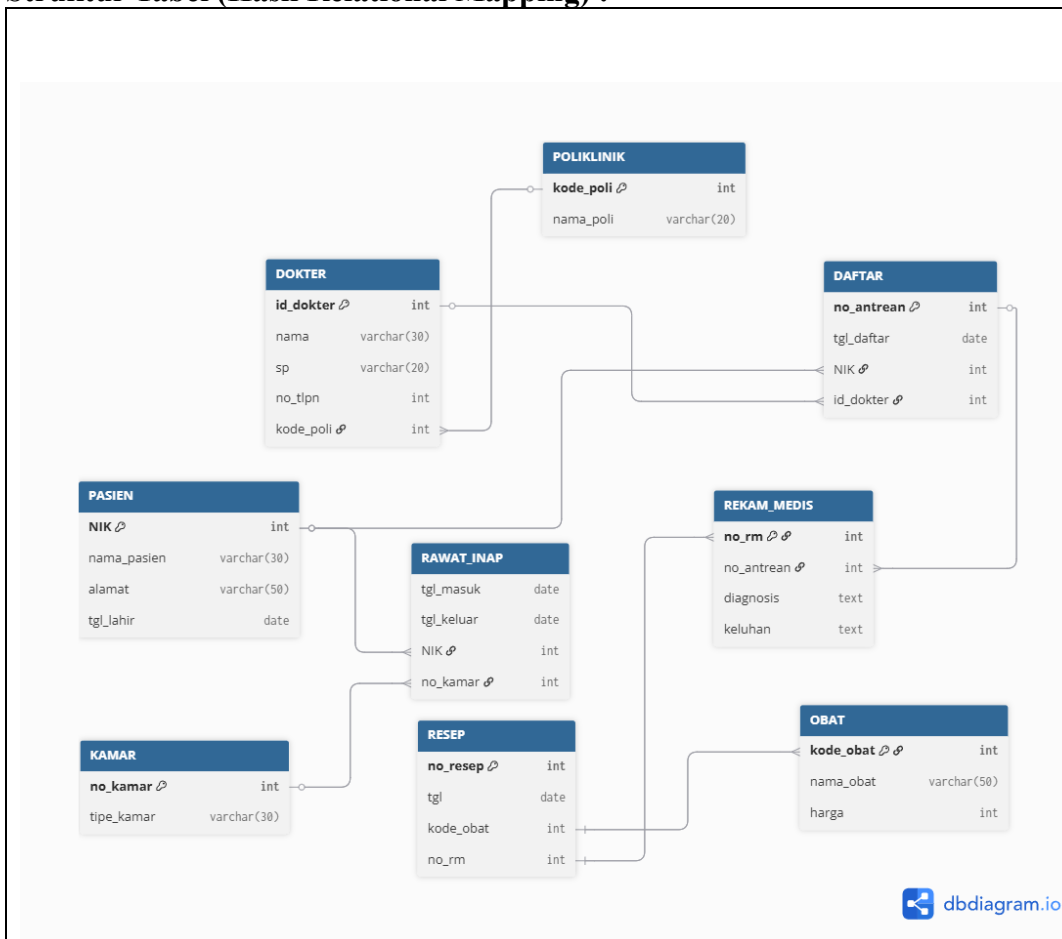
The diagram is an Entity-Relationship (ER) model for a medical system. It features several entities (yellow rectangles) and their attributes (blue ovals). Relationships are indicated by green diamonds.

- pasien** (Patient): Attributes include *nik*, *alamat*, *tipe_kamar*, *tgl_lahir*. Relationships: *tercatat* (with *rekam medis*), *dikonsumsi* (with *obat*), *memeriksa* (with *dokter*), *menempati* (with *kamar*), *menentukan* (with *obat*).
- dokter** (Doctor): Attributes include *id_dokter*, *nama*, *sp*, *nomor_tlpn*, *kode_poli*. Relationships: *memeriksa* (with *pasien*), *poliklinik* (with *poliklinik*).
- obat** (Medicine): Attributes include *kode_obat*, *nama_obat*, *harga*. Relationships: *dikonsumsi* (with *pasien*), *menentukan* (with *pasien*).
- resep** (Prescription): Attributes include *kode_obat*, *nomor_rm*, *tgl*, *no_resep*. Relationship: *tercatat* (with *rekam medis*).
- rekam medis** (Medical Record): Attributes include *diagnosis*, *keluhann*, *nomor_rm*, *nomor_antrian*. Relationship: *tercatat* (with *pasien*).
- kamar** (Room): Attributes include *tipe_kamar*, *nomor_kamar*. Relationship: *menempati* (with *pasien*).
- rawat inap** (Inpatient): Attributes include *nomor_kamar*, *nik*, *tgl_keluar*, *tgl_masuk*. Relationship: *menempati* (with *pasien*).
- poliklinik** (Poliklinik): Attributes include *nama_poli*, *kode_poli*. Relationship: *menentukan* (with *obat*).
- daftar** (Registration): Attributes include *tgl_daftar*, *nik*, *id_dokter*, *nomor_antrian*. Relationship: *memeriksa* (with *dokter*).

Relationships (Green Diamonds):

- tercatat*: Connects *pasien* to *rekam medis* and *resep*.
- dikonsumsi*: Connects *pasien* to *obat*.
- memeriksa*: Connects *pasien* to *dokter*.
- menempati*: Connects *pasien* to *kamar* and *rawat inap*.
- menentukan*: Connects *pasien* to *obat* and *poliklinik*.

Struktur Tabel (Hasil Relational Mapping) :



Dengan menggunakan tiga data berikut :

1. Pelanggan.csv
2. Produk.csv
3. Tr_penjualan.csv

Buat :

1. Database

Query:

```
CREATE DATABASE kelas_c_mart;

USE kelas_c_mart;

CREATE TABLE pelanggan(
kode_pelanggan varchar(20) PRIMARY KEY,
no_urut INT NOT NULL,
nama_pelanggan varchar(30) NOT NULL,
alamat varchar(50)
);

INSERT INTO pelanggan(kode_pelanggan, no_urut,
nama_pelanggan)
VALUES("dqlabcust00", 0, "Pelanggan Non
Member");

INSERT INTO pelanggan(kode_pelanggan, no_urut,
nama_pelanggan, alamat)
VALUES("dqlabcust01", 1, "Eva Novianti, S.H.",
"Vila Sempilan, No. 67 - Kota B"),
("dqlabcust02", 2, "Heidi Goh", "Vila
Sempilan, No. 11 - Kota B"),
("dqlabcust03", 3, "Unang Handoko", "Vila
Sempilan, No. 1 - Kota B");

SELECT * FROM pelanggan;

-- insert data dari file csv
SELECT * FROM kelas_c_mart.ms_pelanggan_dqlab;
```

Output:

	kode_pelanggan	no_urut	nama_pelanggan	alamat
▶	dqlabcust00	0	Pelanggan Non Member	
	dqlabcust01	1	Eva Novianti, S.H.	Vila Sempilan, No. 67 - Kota B
	dqlabcust02	2	Heidi Goh	Vila Sempilan, No. 11 - Kota B
	dqlabcust03	3	Unang Handoko	Vila Sempilan, No. 1 - Kota B
	dqlabcust04	4	Jokolono Sukarman	Vila Permata Intan Berkilau, Blok C5-7
	dqlabcust05	5	Tommy Sinaga	Vila Permata Intan Berkilau, Blok A1/2
	dqlabcust06	6	Irwan Setianto	Vila Gunung Seribu, Blok O1 - No. 1 - Kota C
	dqlabcust07	7	Agus Cahyono	Vila Gunung Seribu, Blok F4 - No. 8
	dqlabcust08	8	Maria Sirait	Vila Bukit Sagitarius, Gang. Sawit No. 3
	dqlabcust09	9	Ir. Ita Nugraha	Vila Bukit Sagitarius, Gang Kelapa No. 6
	dqlabcust10	10	Djoko Wardoyo, Drs.	Vila Bukit Sagitarius, Blok A1 No. 1

2. Tabel

Query:

```
CREATE DATABASE IF NOT EXISTS dqlab_db;
USE dqlab_db;

-- 2. Buat tabel ms_produk_dqlab
CREATE TABLE ms_produk_dqlab (
    kode_produk VARCHAR(10) PRIMARY KEY,
    no_urut INT NOT NULL,
    kategori_produk VARCHAR(50) NOT NULL,
    nama_produk VARCHAR(100) NOT NULL,
    harga DECIMAL(15,2) NOT NULL
);

-- 3. Insert data dari file CSV
SELECT * FROM dqlab_db.ms_produk_dqlab;

-- 4. Verifikasi data (opsional)
SELECT * FROM ms_produk_dqlab ORDER BY no_urut;
```

Output:

	kode_produk	no_urut	kategori_produk	nama_produk	harga
▶	prod-01	1	Alat Tulis Kantor	Kotak Pensil DQLab	62500.00
	prod-02	2	Aksesoris Komputer	Flashdisk DQLab 64 GB	55000.00
	prod-03	3	Gift & Voucher	Gift Voucher DQLab 100rb	100000.00
	prod-04	4	Aksesoris Komputer	Flashdisk DQLab 32 GB	40000.00
	prod-05	5	Gift & Voucher	Gift Voucher DQLab 250rb	250000.00
	prod-06	6	Alat Tulis Kantor	Pulpen Multifunction + Laser DQLab	92500.00
	prod-07	7	Gift & Voucher	Tas Travel Organizer DigiSkills.id	48000.00
	prod-08	8	Gift & Voucher	Gantungan Kunci DQLab	15800.00
	prod-09	9	Alat Tulis Kantor	Buku Planner Agenda DQLab	92000.00
	prod-10	10	Alat Tulis Kantor	Sticky Notes DQLab 500 sheets	55000.00
*	NULL	NULL	NULL	NULL	NULL

3. Isi Data pada tabel

Query:

```
CREATE DATABASE IF NOT EXISTS dqlab_sales;
USE dqlab_sales;

-- Membuat tabel penjualan sesuai struktur CSV
CREATE TABLE tr_penjualan_dqlab (
    kode_urut_transaksi INT PRIMARY KEY,
    kode_transaksi VARCHAR(20) NOT NULL,
    kode_pelanggan VARCHAR(20) NOT NULL,
    tgl_transaksi DATETIME NOT NULL,
    no_urut INT NOT NULL,
    kode_produk VARCHAR(20) NOT NULL,
    nama_produk VARCHAR(100) NOT NULL,
    qty INT NOT NULL,
    harga DECIMAL(15,2) NOT NULL,
    diskon_persen DECIMAL(5,2) NOT NULL DEFAULT 0.00
);

-- Insert data dari file CSV
SELECT * FROM dqlab_sales.tr_penjualan_dqlab;

-- Query verifikasi data
SELECT * FROM tr_penjualan_dqlab ORDER BY
kode_urut_transaksi;
```

Output:

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:										
	kode_urut_transaksi	kode_transaksi	kode_pelanggan	tgl_transaksi	no_urut	kode_produk	nama_produk	qty	harga	diskon_persen
3		tr-001	dqlabcust07	2020-05-01 00:00:00	3	prod-09	Buku Planner Agenda DQSQuad	3	92000.00	0.00
4		tr-001	dqlabcust07	2020-05-01 00:00:00	4	prod-04	Flashdisk DQLab 32 GB	3	40000.00	0.00
5		tr-002	dqlabcust00	2020-05-01 00:00:00	1	prod-03	Gift Voucher DQLab 100rb	2	100000.00	0.00
6		tr-002	dqlabcust00	2020-05-01 00:00:00	2	prod-10	Sticky Notes DQLab 500 sheets	4	55000.00	0.00
7		tr-002	dqlabcust00	2020-05-01 00:00:00	3	prod-07	Tas Travel Organizer DQLab	1	48000.00	0.00
8		tr-003	dqlabcust03	2020-05-03 00:00:00	1	prod-02	Flashdisk DQLab 64 GB	2	55000.00	12.50
9		tr-004	dqlabcust03	2020-05-03 00:00:00	1	prod-10	Sticky Notes DQLab 500 sheets	5	55000.00	0.00
10		tr-004	dqlabcust03	2020-05-03 00:00:00	2	prod-04	Flashdisk DQLab 32 GB	4	40000.00	0.00
11		tr-005	dqlabcust00	2020-06-12 00:00:00	1	prod-09	Buku Planner Agenda DQSQuad	3	92000.00	0.00
12		tr-005	dqlabcust00	2020-06-12 00:00:00	2	prod-01	Kotak Pensil DQLab	1	62500.00	5.00
13		tr-005	dqlabcust00	2020-06-12 00:00:00	3	prod-04	Flashdisk DQLab 32 GB	2	40000.00	0.00
14		tr-006	dqlabcust02	2020-06-19 00:00:00	1	prod-05	Gift Voucher DQLab 250rb	4	250000.00	0.00
15		tr-006	dqlabcust02	2020-06-19 00:00:00	2	prod-08	Gantungan Kunci DQLab	2	15800.00	0.00
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

tr_penjualan_dqlab 13 x Apply

MODUL 2

SELECT, PREFIX, DAN ALIAS

Penjelasan laporan:

Dokumentasikan seluruh aktivitas praktikum untuk modul 2 dengan menuliskan query dan menempelkan screenshot output dari query tersebut.

Dengan ketentuan:

1. Gunakan font `courier new` untuk menuliskan query
2. Tempelkan Screenshot hasil query dalam kolom output

Praktek 1

Perintah SELECT: Mengambil Satu Kolom

Query:

```
-- menggunakan database
USE kelas_c_mart;

SELECT kode_pelanggan FROM pelanggan;
```

Output:



The screenshot shows a 'Result Grid' window with a 'Filter Rows' field. The grid displays the results of the query 'SELECT kode_pelanggan FROM pelanggan;'. The first column is labeled 'kode_pelanggan'. There are four data rows with values 'dqlabcust00', 'dqlabcust01', 'dqlabcust02', and 'dqlabcust03'. A fifth row is labeled 'NULL' with a star icon to its left.

kode_pelanggan
dqlabcust00
dqlabcust01
dqlabcust02
dqlabcust03
NULL

Praktek 2

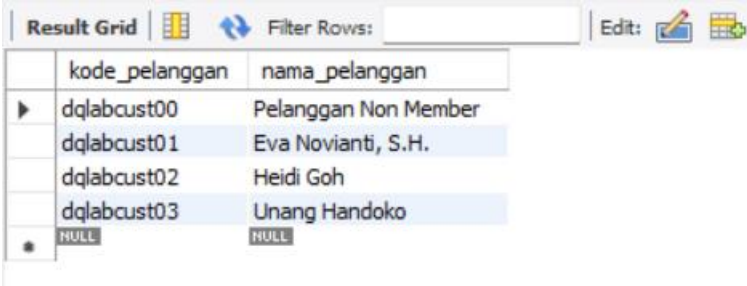
Perintah SELECT: Mengambil lebih dari satu kolom

Query:

```
-- menggunakan database
USE kelas_c_mart;

SELECT kode_pelanggan, nama_pelanggan FROM pelanggan;
```

Output:



The screenshot shows a database query result grid with two columns: 'kode_pelanggan' and 'nama_pelanggan'. The data is as follows:

kode_pelanggan	nama_pelanggan
dqlabcust00	Pelanggan Non Member
dqlabcust01	Eva Novianti, S.H.
dqlabcust02	Heidi Goh
dqlabcust03	Unang Handoko
NULL	NULL

Praktek 3

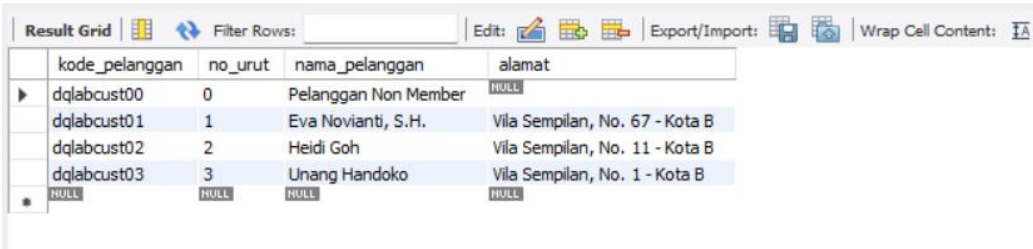
Perintah SELECT: Mengambil Seluruh Kolom

Query:

```
-- menggunakan database
USE kelas_c_mart;

SELECT * FROM pelanggan;
```

Output:



The screenshot shows a database query result grid with four columns: 'kode_pelanggan', 'no_urut', 'nama_pelanggan', and 'alamat'. The data is as follows:

kode_pelanggan	no_urut	nama_pelanggan	alamat
dqlabcust00	0	Pelanggan Non Member	NULL
dqlabcust01	1	Eva Novianti, S.H.	Vila Sempilan, No. 67 - Kota B
dqlabcust02	2	Heidi Goh	Vila Sempilan, No. 11 - Kota B
dqlabcust03	3	Unang Handoko	Vila Sempilan, No. 1 - Kota B
NULL	NULL	NULL	NULL

Latihan Mandiri

Soal 1

Query:

```
-- menggunakan database
USE dqlab_db;

SELECT kode_produk, nama_produk FROM ms_produk_dqlab;
```


Praktek 4

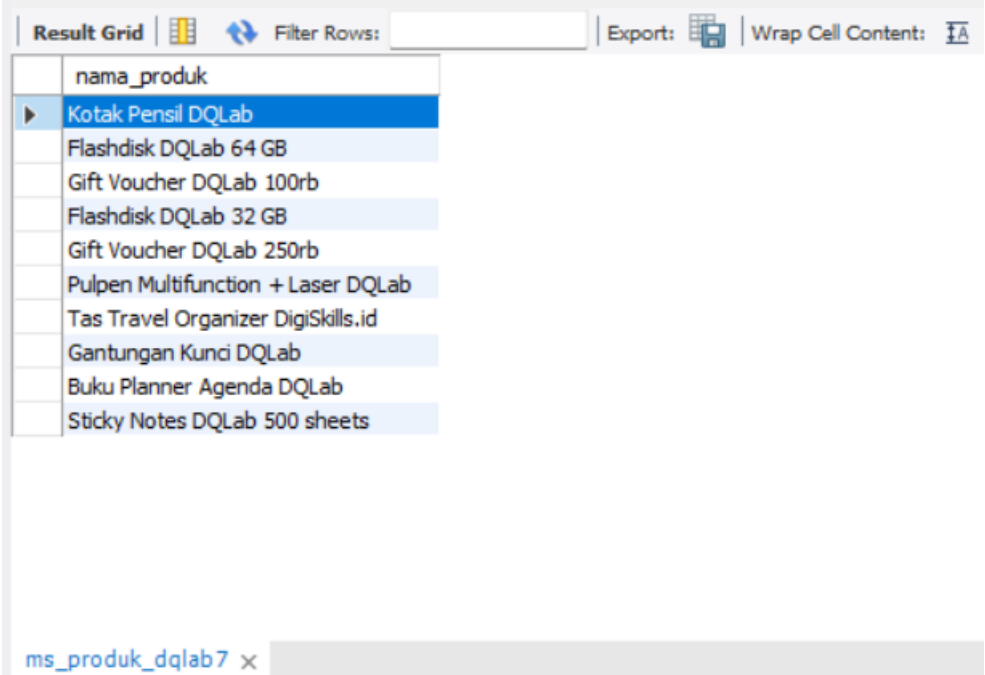
PREFIX: berupa nama table

Query:

```
-- menggunakan database
USE dqlab_db;

SELECT ms_produk_dqlab.nama_produk FROM
ms_produk_dqlab;
```

Output:



The screenshot shows a database query result grid. At the top, there is a toolbar with options like 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. Below the toolbar, a table is displayed with a single column named 'nama_produk'. The table contains 12 rows of product names. The first row, 'Kotak Pensil DQLab', is highlighted in blue. At the bottom of the grid, there is a tab labeled 'ms_produk_dqlab7' with a close button (x).

nama_produk
Kotak Pensil DQLab
Flashdisk DQLab 64 GB
Gift Voucher DQLab 100rb
Flashdisk DQLab 32 GB
Gift Voucher DQLab 250rb
Pulpen Multifunction + Laser DQLab
Tas Travel Organizer DigiSkills.id
Gantungan Kunci DQLab
Buku Planner Agenda DQLab
Sticky Notes DQLab 500 sheets

Praktek 5

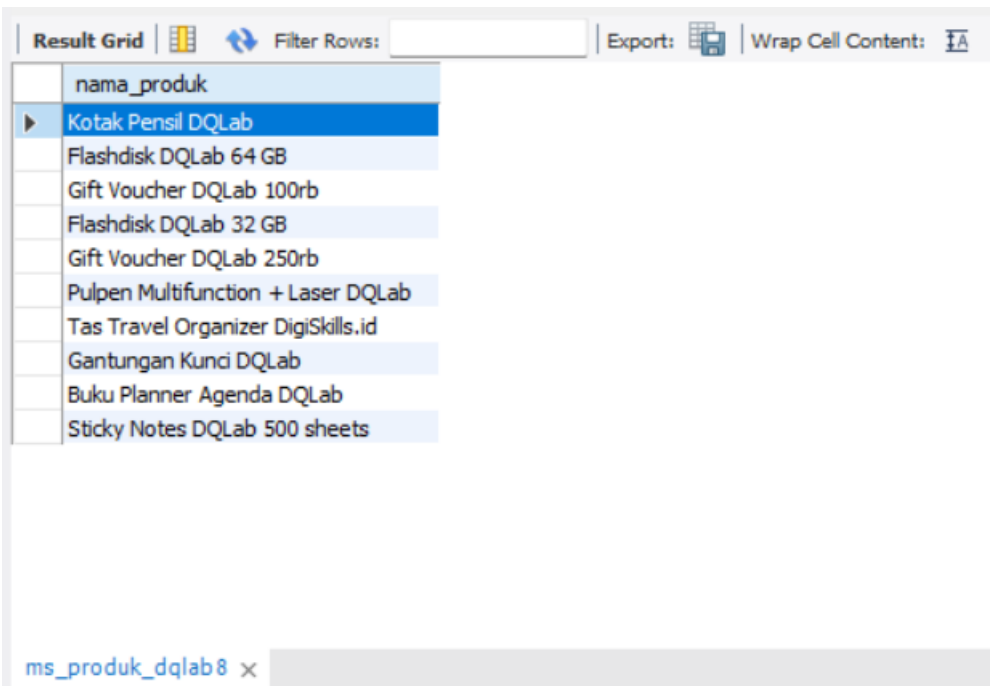
PREFIX: berupa nama database dan tabel

Query:

```
-- menggunakan database
USE dqlab_db;

SELECT dqlab_db.ms_produk_dqlab.nama_produk FROM
ms_produk_dqlab;
```

Output:



nama_produk
Kotak Pensil DQLab
Flashdisk DQLab 64 GB
Gift Voucher DQLab 100rb
Flashdisk DQLab 32 GB
Gift Voucher DQLab 250rb
Pulpen Multifunction + Laser DQLab
Tas Travel Organizer DigiSkills.id
Gantungan Kunci DQLab
Buku Planner Agenda DQLab
Sticky Notes DQLab 500 sheets

ms_produk_dqlab8 x

Praktek 6

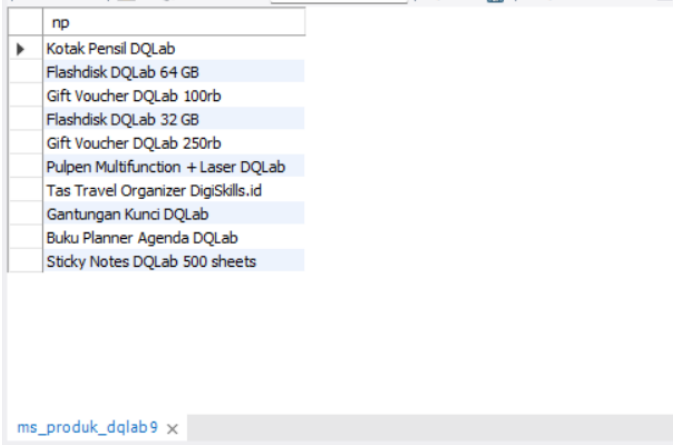
ALIAS: pada kolom

Query:

```
-- menggunakan database
USE dqlab_db;

SELECT nama_produk as np FROM ms_produk_dqlab;
```

Output:



np
Kotak Pensil DQLab
Flashdisk DQLab 64 GB
Gift Voucher DQLab 100rb
Flashdisk DQLab 32 GB
Gift Voucher DQLab 250rb
Pulpen Multifunction + Laser DQLab
Tas Travel Organizer DigiSkills.id
Gantungan Kunci DQLab
Buku Planner Agenda DQLab
Sticky Notes DQLab 500 sheets

ms_produk_dqlab9 x

Praktek 7

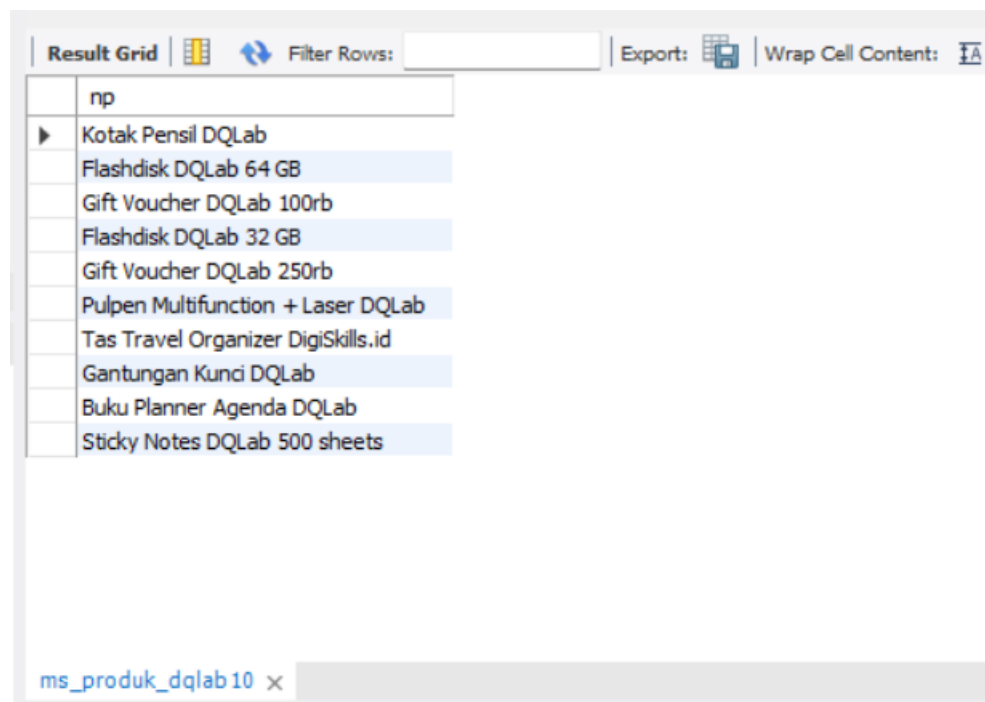
ALIAS: tanpa keyword AS

Query:

```
-- menggunakan database
USE dqlab_db;

SELECT nama_produk np FROM ms_produk_dqlab;
```

Output:



The screenshot shows a database query result grid. The grid has a header row with the column name 'np'. Below the header, there is a list of 12 products. The products are: Kotak Pensil DQLab, Flashdisk DQLab 64 GB, Gift Voucher DQLab 100rb, Flashdisk DQLab 32 GB, Gift Voucher DQLab 250rb, Pulpen Multifunction + Laser DQLab, Tas Travel Organizer DigiSkills.id, Gantungan Kunci DQLab, Buku Planner Agenda DQLab, and Sticky Notes DQLab 500 sheets. The table is titled 'ms_produk_dqlab 10 x'.

np
Kotak Pensil DQLab
Flashdisk DQLab 64 GB
Gift Voucher DQLab 100rb
Flashdisk DQLab 32 GB
Gift Voucher DQLab 250rb
Pulpen Multifunction + Laser DQLab
Tas Travel Organizer DigiSkills.id
Gantungan Kunci DQLab
Buku Planner Agenda DQLab
Sticky Notes DQLab 500 sheets

Praktek 8

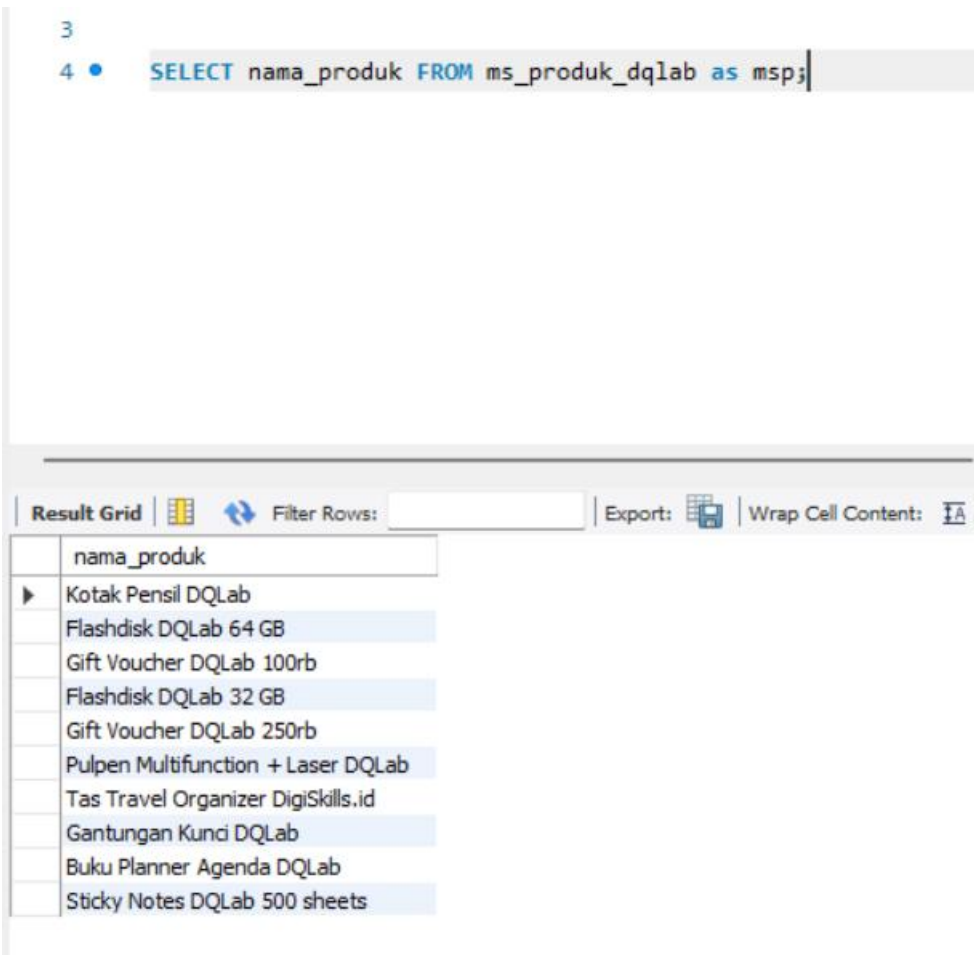
ALIAS: Pada nama table

Query:

```
-- menggunakan database
USE dqlab_db;

SELECT nama_produk FROM ms_produk_dqlab as msp;
```

Output:



The screenshot shows a database query interface. At the top, a SQL query is entered in a text area:

```
3  
4 • SELECT nama_produk FROM ms_produk_dqlab as msp;
```

Below the query area, there is a toolbar with options: "Result Grid", "Filter Rows:", "Export:", and "Wrap Cell Content:". The "Result Grid" option is selected, and a table of results is displayed below it.

nama_produk
Kotak Pensil DQLab
Flashdisk DQLab 64 GB
Gift Voucher DQLab 100rb
Flashdisk DQLab 32 GB
Gift Voucher DQLab 250rb
Pulpen Multifunction + Laser DQLab
Tas Travel Organizer DigiSkills.id
Gantungan Kunci DQLab
Buku Planner Agenda DQLab
Sticky Notes DQLab 500 sheets

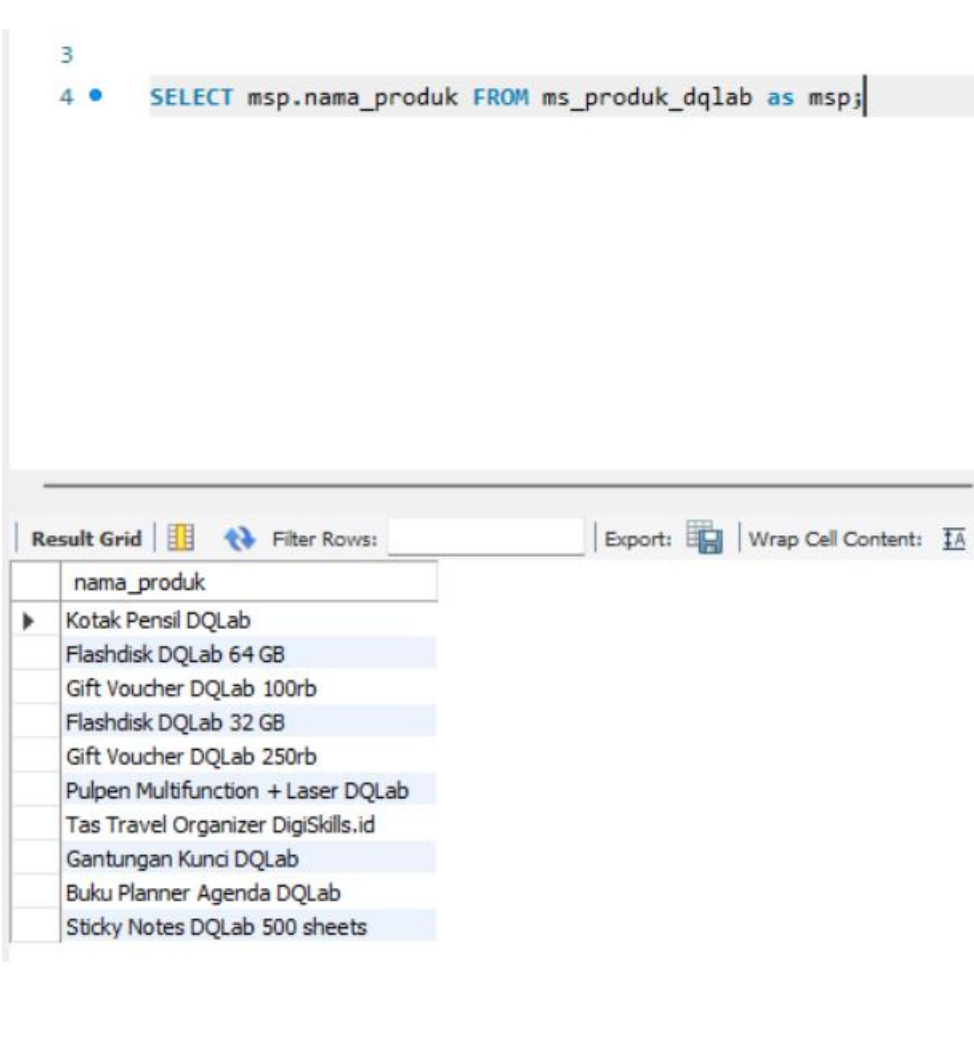
Praktek 9

Penggunaan PREFIX dan ALIAS pada Nama Table

Query:

```
-- menggunakan database  
USE dqlab_db;  
  
SELECT msp.nama_produk FROM ms_produk_dqlab as msp;
```

Output:



The screenshot shows a SQL query execution interface. At the top, a query is entered: `SELECT msp.nama_produk FROM ms_produk_dqlab as msp;`. Below the query, the results are displayed in a table. The table has one column, `nama_produk`, and ten rows of product names. The interface includes a 'Result Grid' tab, a 'Filter Rows' input field, and buttons for 'Export' and 'Wrap Cell Content'.

nama_produk
Kotak Pensil DQLab
Flashdisk DQLab 64 GB
Gift Voucher DQLab 100rb
Flashdisk DQLab 32 GB
Gift Voucher DQLab 250rb
Pulpen Multifunction + Laser DQLab
Tas Travel Organizer DigiSkills.id
Gantungan Kunci DQLab
Buku Planner Agenda DQLab
Sticky Notes DQLab 500 sheets

Case 1

Tim sales ini mencari list nama customer dari database, berupa nama customer nya serta alamat nya dimana marketing ingin mencoba melakukan follow up penjualan produk dengan melakukan kunjungan ke alamat customer tersebut

Query:

```
-- menggunakan database
USE kelas_c_mart;

SELECT nama_pelanggan, alamat FROM pelanggan;
```


Output:

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	nama_pelanggan	alamat		
▶	Pelanggan Non Member	NULL		
	Eva Novianti, S.H.	Vila Sempilan, No. 67 - Kota B		
	Heidi Goh	Vila Sempilan, No. 11 - Kota B		
	Unang Handoko	Vila Sempilan, No. 1 - Kota B		

pelanggan 13 ×

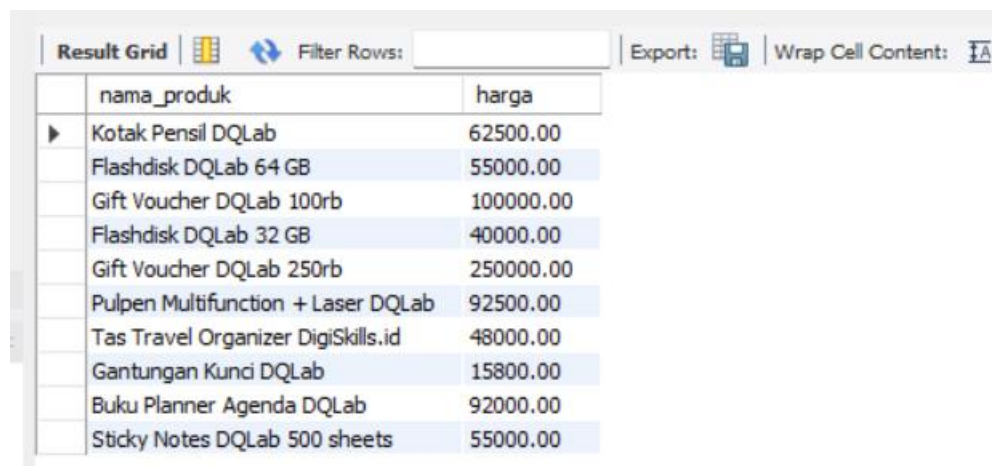
Case 2

Sales dan Marketing ingin melihat nama produk dan harga dari setiap produk tersebut untuk membuat strategi penjualan ke konsumen berdasarkan harga produk tersebut.

Query:

```
-- menggunakan database  
USE dqlab_db;  
  
SELECT nama_produk, harga FROM ms_produk_dqlab;
```

Output:



The screenshot shows a database query result grid with two columns: 'nama_produk' and 'harga'. The grid contains 11 rows of data. The interface includes a 'Result Grid' tab, a 'Filter Rows' search bar, and 'Export' and 'Wrap Cell Content' buttons.

	nama_produk	harga
▶	Kotak Pensil DQLab	62500.00
	Flashdisk DQLab 64 GB	55000.00
	Gift Voucher DQLab 100rb	100000.00
	Flashdisk DQLab 32 GB	40000.00
	Gift Voucher DQLab 250rb	250000.00
	Pulpen Multifunction + Laser DQLab	92500.00
	Tas Travel Organizer DigiSkills.id	48000.00
	Gantungan Kunci DQLab	15800.00
	Buku Planner Agenda DQLab	92000.00
	Sticky Notes DQLab 500 sheets	55000.00

MODUL 3

NILAI LITERAL, OPERATOR MATEMATIKA, OPERATOR PERBANDINGAN, FUNGSI MATEMATIKA, FUNGSI TANGGAL, KLAUSUL WHERE, OPERATOR LIKE, OPERATOR LOGIKA

Penjelasan laporan:

Dokumentasikan seluruh aktivitas praktikum untuk modul 2 dengan menuliskan query dan menempelkan screenshot output dari query tersebut.

Dengan ketentuan:

1. Gunakan font `courier new` untuk menuliskan query
2. Tempelkan Screenshot hasil query dalam kolom output

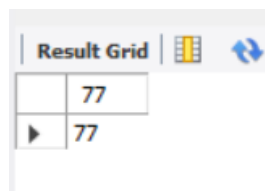
Praktek 1

SELECT dengan satu nilai literal

Query:

```
SELECT 77;
```

Output:



Result Grid	
	77

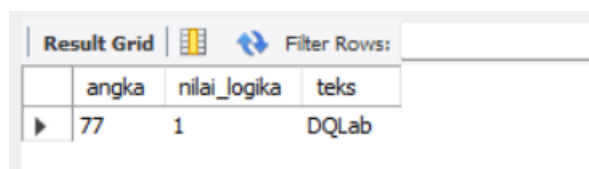
Praktek 2

SELECT dengan beberapa nilai literal

Query:

```
SELECT 77 AS angka, true AS nilai_logika, 'DQLab' AS teks;
```

Output:



Result Grid			
	angka	nilai_logika	teks
	77	1	DQLab

Praktek 3

SELECT dengan literal NULL

Query:

```
SELECT NULL AS kosong;
```

Output:



	kosong
▶	NULL

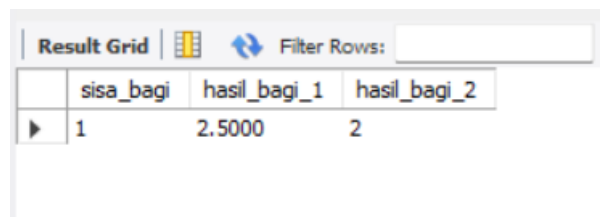
Praktek 4

SELECT dengan perhitungan matematika

Query:

```
SELECT  
5%2 AS sisa_bagi,  
5/2 AS hasil_bagi_1,  
5 DIV 2 AS hasil_bagi_2;
```

Output:



	sisa_bagi	hasil_bagi_1	hasil_bagi_2
▶	1	2.5000	2

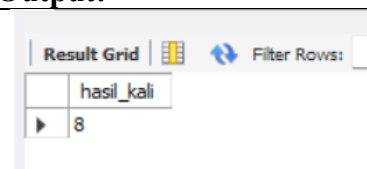
Latihan Mandiri

Soal 1

Query:

```
SELECT  
4*2 AS hasil_kali;
```

Output:



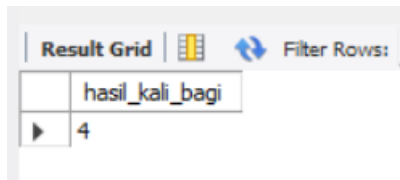
	hasil_kali
▶	8

Soal 2

Query:

```
SELECT  
(4*8)%7 AS hasil_kali_bagi;
```

Output:



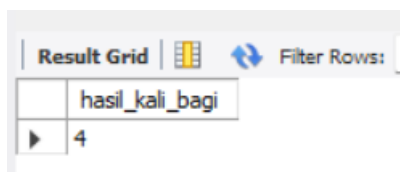
	hasil_kali_bagi
▶	4

Soal 3

Query:

```
SELECT  
(4*8) MOD 7 AS hasil_kali_bagi;  
  
-- MOD adalah alternatif dari %
```

Output:



	hasil_kali_bagi
▶	4

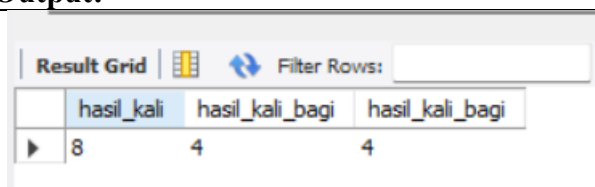
Soal 4

Gabungkan soal 1 sd 3 dalam 1 query

Query:

```
SELECT  
4*2 as hasil_kali,  
(4*8) % 7 as hasil_kali_bagi,  
(4*8) MOD 7 AS hasil_kali_bagi;  
  
-- MOD adalah alternatif dari %
```

Output:



	hasil_kali	hasil_kali_bagi	hasil_kali_bagi
▶	8	4	4

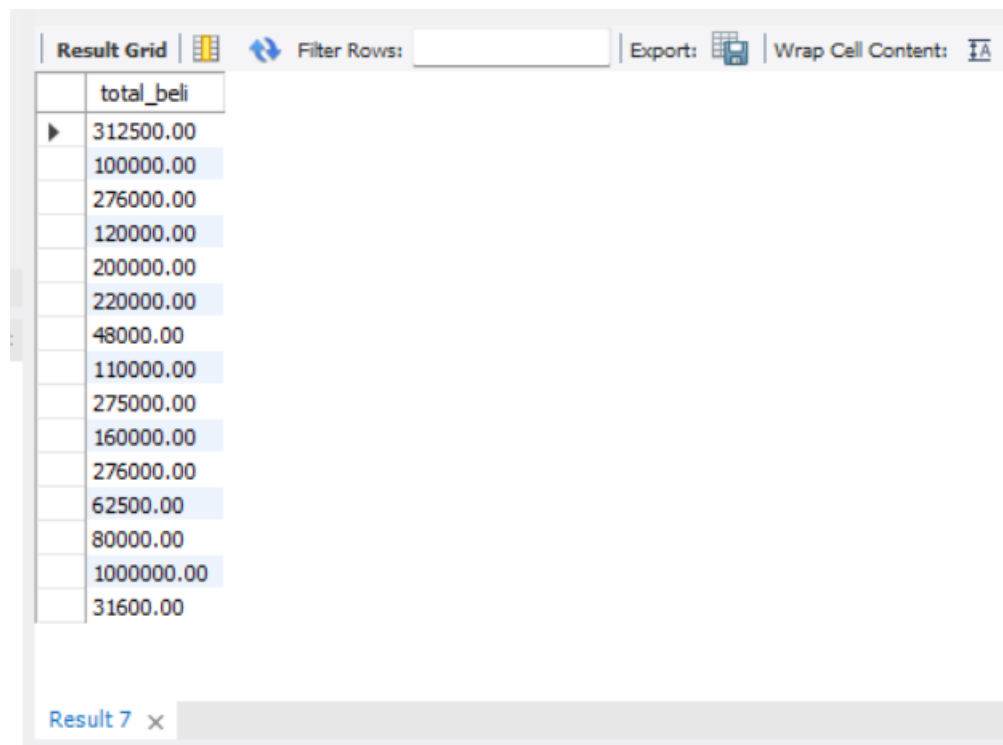
Praktek 5

Ekspresi Matematika pada kolom table

Query:

```
-- menggunakan database:  
USE dqlab_sales;  
  
SELECT qty*harga AS total_beli FROM  
tr_penjualan_dqlab;
```

Output:



The screenshot shows a 'Result Grid' window with a toolbar at the top containing 'Filter Rows', 'Export', and 'Wrap Cell Content' options. The grid displays a single column named 'total_beli' with 16 rows of numerical data. The values are: 312500.00, 100000.00, 276000.00, 120000.00, 200000.00, 220000.00, 48000.00, 110000.00, 275000.00, 160000.00, 276000.00, 62500.00, 80000.00, 1000000.00, and 31600.00. The status bar at the bottom indicates 'Result 7'.

total_beli
312500.00
100000.00
276000.00
120000.00
200000.00
220000.00
48000.00
110000.00
275000.00
160000.00
276000.00
62500.00
80000.00
1000000.00
31600.00

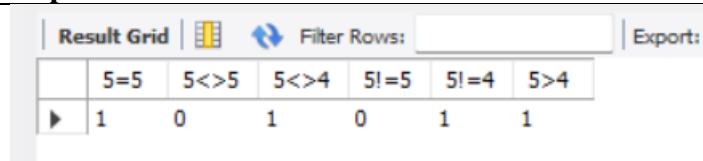
Praktek 6

SELECT dengan ekspresi perbandingan

Query:

```
SELECT 5=5, 5<>5, 5<>4, 5!=5, 5!=4, 5>4;
```

Output:



The screenshot shows a 'Result Grid' window with a toolbar at the top containing 'Filter Rows' and 'Export' options. The grid displays six columns representing comparison expressions: '5=5', '5<>5', '5<>4', '5!=5', '5!=4', and '5>4'. The first row of data shows the results for these expressions: 1, 0, 1, 0, 1, and 1 respectively.

5=5	5<>5	5<>4	5!=5	5!=4	5>4
1	0	1	0	1	1

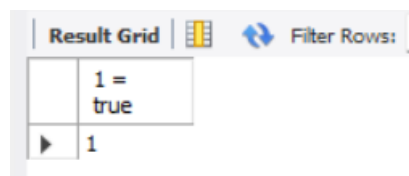
Latihan Mandiri

Soal 1

Query:

```
SELECT 1 = true;
```

Output:



The screenshot shows a 'Result Grid' with a single row containing the value 'true'. Below the grid, a 'Filter Rows:' button is visible. The row is highlighted with a blue background.

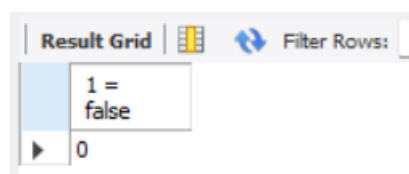
1 =	true
-----	------

Soal 2

Query:

```
SELECT 1 = false;
```

Output:



The screenshot shows a 'Result Grid' with a single row containing the value 'false'. Below the grid, a 'Filter Rows:' button is visible. The row is highlighted with a blue background.

1 =	false
-----	-------

Soal 3

Query:

```
SELECT 5>=5;
```

Output:



The screenshot shows a 'Result Grid' with a single row containing the value '1'. Below the grid, a 'Filter Rows:' button is visible. The row is highlighted with a blue background.

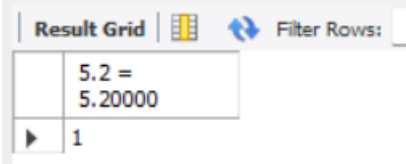
5>=5	1
------	---

Soal 4

Query:

```
SELECT 5.2 = 5.20000;
```

Output:



The screenshot shows a SQL query result grid. The top bar has 'Result Grid', a grid icon, a refresh icon, and 'Filter Rows:'. The grid has one row with the text '5.2 =' in the first column and '5.20000' in the second column. Below the grid is a scroll bar with a right arrow and the number '1'.

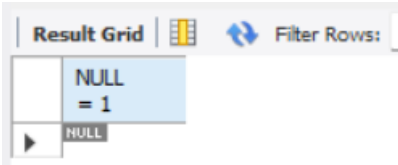
5.2 =	5.20000
-------	---------

Soal 5

Query:

```
SELECT NULL = 1;
```

Output:



The screenshot shows a SQL query result grid. The top bar has 'Result Grid', a grid icon, a refresh icon, and 'Filter Rows:'. The grid has one row with the text 'NULL =' in the first column and '1' in the second column. Below the grid is a scroll bar with a right arrow and the text 'NULL'.

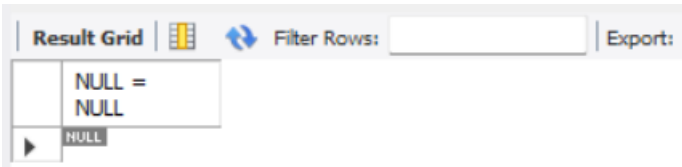
NULL =	1
--------	---

Soal 6

Query:

```
SELECT NULL = NULL;
```

Output:



The screenshot shows a SQL query result grid. The top bar has 'Result Grid', a grid icon, a refresh icon, 'Filter Rows:', an empty text box, and 'Export:'. The grid has one row with the text 'NULL =' in the first column and 'NULL' in the second column. Below the grid is a scroll bar with a right arrow and the text 'NULL'.

NULL =	NULL
--------	------

Soal 7

Gabungkan soal 1 sd 6 dalam 1 query

Query:

```
SELECT 1 = true,  
1 = false,  
5>=5,  
5.2 = 5.20000,  
NULL = 1,  
NULL = NULL;
```


Output:

	1 = true	1 = false	5 >= 5	5.2 = 5.20000	NULL = 1	NULL = NULL
▶	1	0	1	1	NULL	NULL

Soal 8

Dari perbandingan di atas, manakah yang tidak menghasilkan nilai 1 atau 0?

Ekspresi yang tidak menghasilkan nilai 1 dan 0 adalah NULL=1 dan NULL=NULL, Karena keduanya menghasilkan nilai "NULL"

Praktek 7

Perbandingan pada kolom table

Query:

```
USE dqlab_sales;  
  
SELECT qty > 3 from tr_penjualan_dqlab;
```

Output:

	qty > 3
▶	1
	0
	0
	0
	0
	1
	0
	0
	1
	1
	0
	0
	0
	1
	0

tr_penjualan_dqlab 23 x

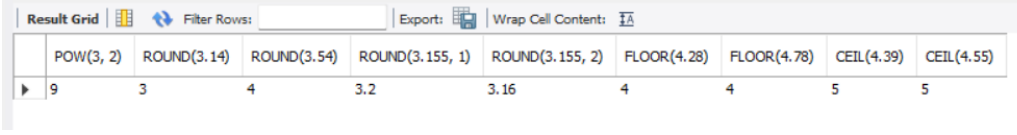
Praktek 8

Penggunaan fungsi dengan nilai literal

Query:

```
SELECT POW(3, 2), ROUND(3.14), ROUND(3.54),  
       ROUND(3.155, 1), ROUND(3.155, 2),  
       FLOOR(4.28), FLOOR(4.78),  
       CEIL(4.39), CEIL(4.55);
```

Output:



	POW(3, 2)	ROUND(3.14)	ROUND(3.54)	ROUND(3.155, 1)	ROUND(3.155, 2)	FLOOR(4.28)	FLOOR(4.78)	CEIL(4.39)	CEIL(4.55)
▶	9	3	4	3.2	3.16	4	4	5	5

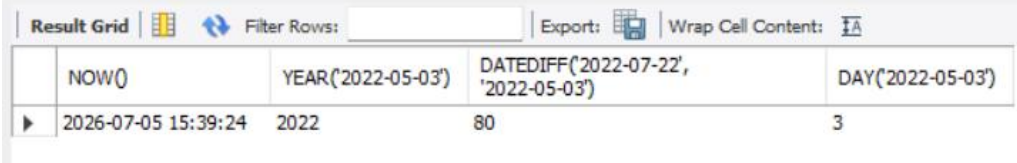
Praktek 9

Penggunaan fungsi tanggal pada literal

Query:

```
SELECT NOW(), YEAR('2022-05-03'), DATEDIFF('2022-07-22',  
       '2022-05-03'), DAY('2022-05-03');
```

Output:



	NOW()	YEAR('2022-05-03')	DATEDIFF('2022-07-22', '2022-05-03')	DAY('2022-05-03')
▶	2026-07-05 15:39:24	2022	80	3

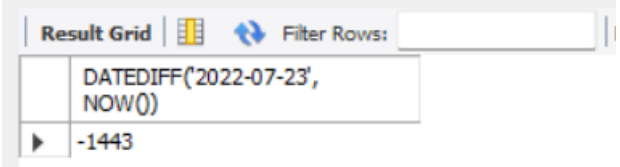
Latihan Mandiri

Soal 1

Query:

```
SELECT DATEDIFF('2022-07-23', NOW());
```

Output:



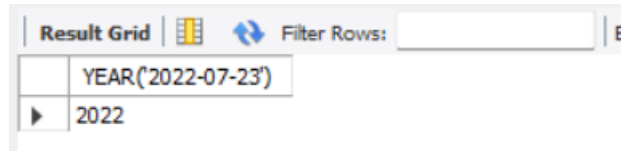
	DATEDIFF('2022-07-23', NOW())
▶	-1443

Soal 2

Query:

```
SELECT YEAR('2022-07-23');
```

Output:



The screenshot shows a database interface with a 'Result Grid' tab. The grid has two columns: the first column contains the SQL expression 'YEAR('2022-07-23')' and the second column contains the result '2022'. There is a 'Filter Rows' search bar to the right of the grid.

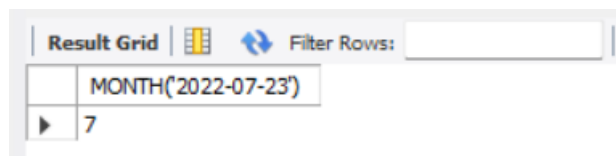
	YEAR('2022-07-23')
▶	2022

Soal 3

Query:

```
SELECT MONTH('2022-07-23');
```

Output:



The screenshot shows a database interface with a 'Result Grid' tab. The grid has two columns: the first column contains the SQL expression 'MONTH('2022-07-23')' and the second column contains the result '7'. There is a 'Filter Rows' search bar to the right of the grid.

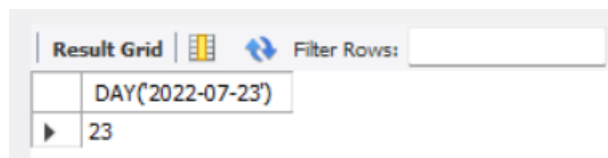
	MONTH('2022-07-23')
▶	7

Soal 4

Query:

```
SELECT DAY('2022-07-23');
```

Output:



The screenshot shows a database interface with a 'Result Grid' tab. The grid has two columns: the first column contains the SQL expression 'DAY('2022-07-23')' and the second column contains the result '23'. There is a 'Filter Rows' search bar to the right of the grid.

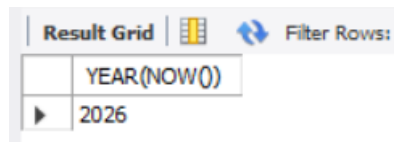
	DAY('2022-07-23')
▶	23

Soal 5

Query:

```
SELECT YEAR(NOW());
```

Output:



Result Grid	
	YEAR(NOW())
▶	2026

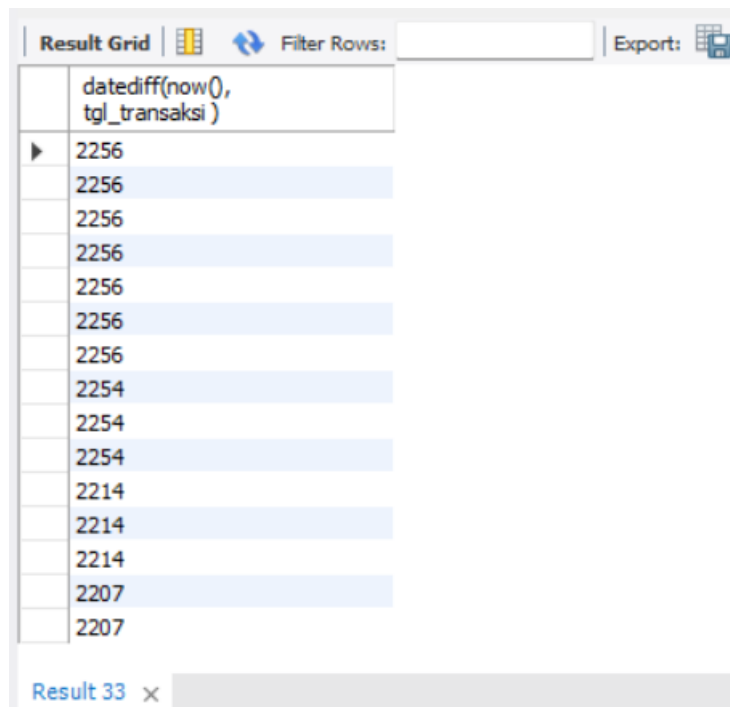
Praktek 10

Penggunaan fungsi tanggal pada kolom table

Query:

```
USE dqlab_sales;  
  
SELECT datediff(now(), tgl_transaksi ) from  
tr_penjualan_dqlab;
```

Output:



Result Grid	
	datediff(now(), tgl_transaksi)
▶	2256
	2256
	2256
	2256
	2256
	2256
	2256
	2254
	2254
	2254
	2214
	2214
	2214
	2207
	2207

Result 33 x

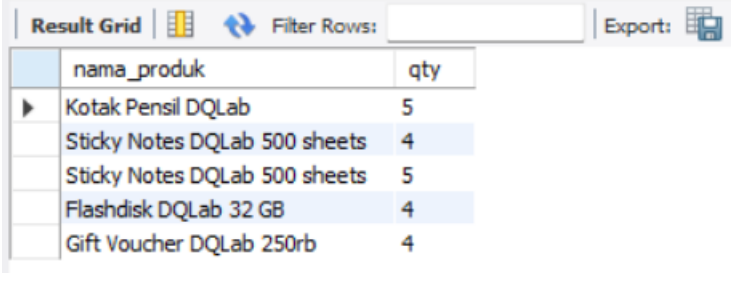
Praktek 11

Penggunaan WHERE

Query:

```
USE dqlab_sales;  
  
SELECT nama_produk, qty FROM tr_penjualan_dqlab WHERE  
qty>3;
```

Output:



The screenshot shows a database interface with a 'Result Grid' tab. It contains a table with two columns: 'nama_produk' and 'qty'. The data rows are: Kotak Pensil DQLab (5), Sticky Notes DQLab 500 sheets (4), Sticky Notes DQLab 500 sheets (5), Flashdisk DQLab 32 GB (4), and Gift Voucher DQLab 250rb (4). There are also icons for 'Filter Rows' and 'Export'.

nama_produk	qty
Kotak Pensil DQLab	5
Sticky Notes DQLab 500 sheets	4
Sticky Notes DQLab 500 sheets	5
Flashdisk DQLab 32 GB	4
Gift Voucher DQLab 250rb	4

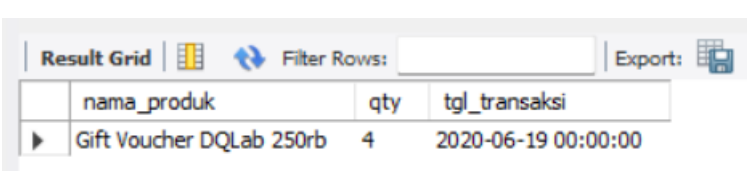
Praktek 12

Penggunaan WHERE dengan multi kriteria

Query:

```
USE dqlab_sales;  
  
SELECT nama_produk, qty, tgl_transaksi  
FROM tr_penjualan_dqlab WHERE qty>3 AND  
month(tgl_transaksi)=6;
```

Output:



The screenshot shows a database interface with a 'Result Grid' tab. It contains a table with three columns: 'nama_produk', 'qty', and 'tgl_transaksi'. The data row is: Gift Voucher DQLab 250rb (4) on 2020-06-19 00:00:00. There are also icons for 'Filter Rows' and 'Export'.

nama_produk	qty	tgl_transaksi
Gift Voucher DQLab 250rb	4	2020-06-19 00:00:00

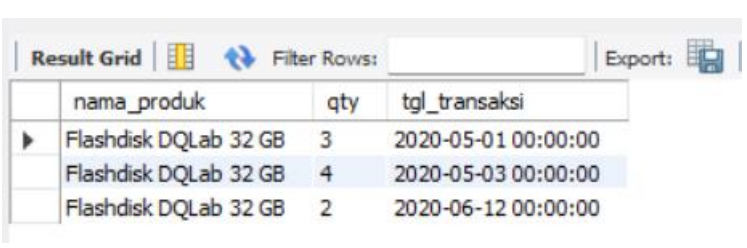
Praktek 13

Penggunaan WHERE dengan kriteria teks

Query:

```
USE dqlab_sales;  
  
SELECT nama_produk, qty, tgl_transaksi  
FROM tr_penjualan_dqlab WHERE nama_produk =  
'Flashdisk DQLab 32 GB';
```

Output:



The screenshot shows a database interface with a 'Result Grid' tab. It contains a table with three columns: 'nama_produk', 'qty', and 'tgl_transaksi'. The data rows are: Flashdisk DQLab 32 GB (3) on 2020-05-01 00:00:00, Flashdisk DQLab 32 GB (4) on 2020-05-03 00:00:00, and Flashdisk DQLab 32 GB (2) on 2020-06-12 00:00:00. There are also icons for 'Filter Rows' and 'Export'.

nama_produk	qty	tgl_transaksi
Flashdisk DQLab 32 GB	3	2020-05-01 00:00:00
Flashdisk DQLab 32 GB	4	2020-05-03 00:00:00
Flashdisk DQLab 32 GB	2	2020-06-12 00:00:00

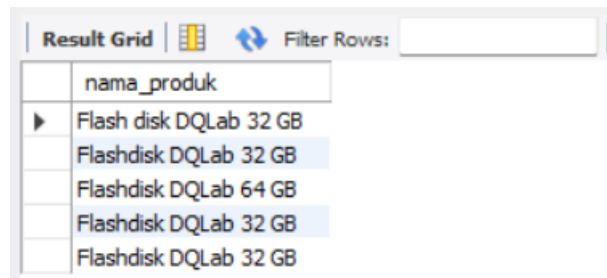
Praktek 14

Penggunaan LIKE

Query:

```
USE dqlab_sales;  
  
SELECT nama_produk FROM tr_penjualan_dqlab WHERE  
nama_produk LIKE 'f%';
```

Output:



The screenshot shows a SQL query result grid with the following data:

	nama_produk
▶	Flash disk DQLab 32 GB
	Flashdisk DQLab 32 GB
	Flashdisk DQLab 64 GB
	Flashdisk DQLab 32 GB
	Flashdisk DQLab 32 GB

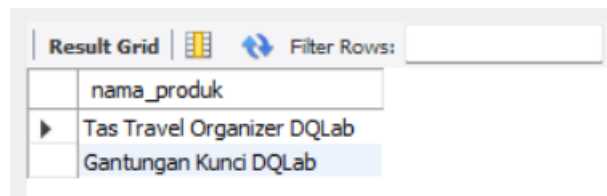
Latihan Mandiri

Soal 1

Query:

```
USE dqlab_sales;  
  
SELECT nama_produk FROM tr_penjualan_dqlab WHERE  
nama_produk LIKE '_a%';
```

Output:



The screenshot shows a SQL query result grid with the following data:

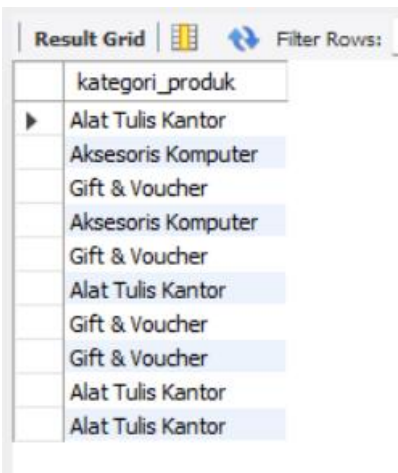
	nama_produk
▶	Tas Travel Organizer DQLab
	Gantungan Kunci DQLab

Soal 2

Query:

```
USE dqlab_db;  
SELECT kategori_produk FROM ms_produk_dqlab WHERE  
kategori_produk LIKE '%t%';
```

Output:



The screenshot shows a database query result grid with a header row and ten data rows. The header row is labeled 'kategori_produk'. The data rows contain the following values: 'Alat Tulis Kantor', 'Aksesoris Komputer', 'Gift & Voucher', 'Aksesoris Komputer', 'Gift & Voucher', 'Alat Tulis Kantor', 'Gift & Voucher', 'Gift & Voucher', 'Alat Tulis Kantor', and 'Alat Tulis Kantor'. The grid has a 'Filter Rows' button and a 'Result Grid' tab.

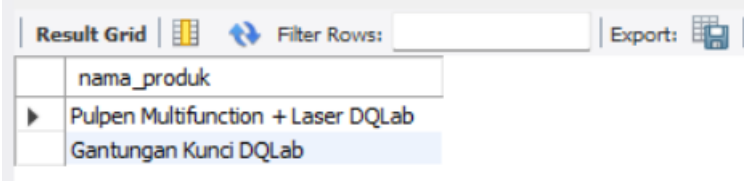
kategori_produk
Alat Tulis Kantor
Aksesoris Komputer
Gift & Voucher
Aksesoris Komputer
Gift & Voucher
Alat Tulis Kantor
Gift & Voucher
Gift & Voucher
Alat Tulis Kantor
Alat Tulis Kantor

Soal 3

Query:

```
USE dqlab_db;
SELECT nama_produk FROM ms_produk_dqlab WHERE
nama_produk LIKE '%un%';
```

Output:



The screenshot shows a database query result grid with a header row and two data rows. The header row is labeled 'nama_produk'. The data rows contain the following values: 'Pulpen Multifunction + Laser DQLab' and 'Gantungan Kunci DQLab'. The grid has a 'Filter Rows' button, an 'Export' button, and a 'Result Grid' tab.

nama_produk
Pulpen Multifunction + Laser DQLab
Gantungan Kunci DQLab

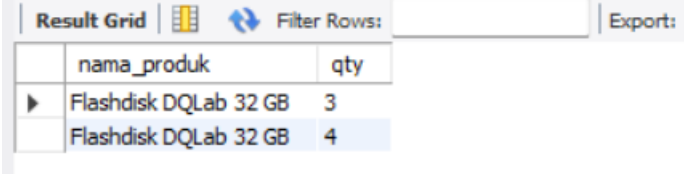
Praktek 15

Filtering dengan operator logika AND

Query:

```
USE dqlab_sales;
SELECT nama_produk, qty FROM tr_penjualan_dqlab WHERE
nama_produk LIKE 'f%' AND qty > 2;
```

Output:



The screenshot shows a database query result grid with a header row and two data rows. The header row has two columns: 'nama_produk' and 'qty'. The data rows contain the following values: 'Flashdisk DQLab 32 GB' with a quantity of 3, and 'Flashdisk DQLab 32 GB' with a quantity of 4. The grid has a 'Filter Rows' button, an 'Export' button, and a 'Result Grid' tab.

nama_produk	qty
Flashdisk DQLab 32 GB	3
Flashdisk DQLab 32 GB	4

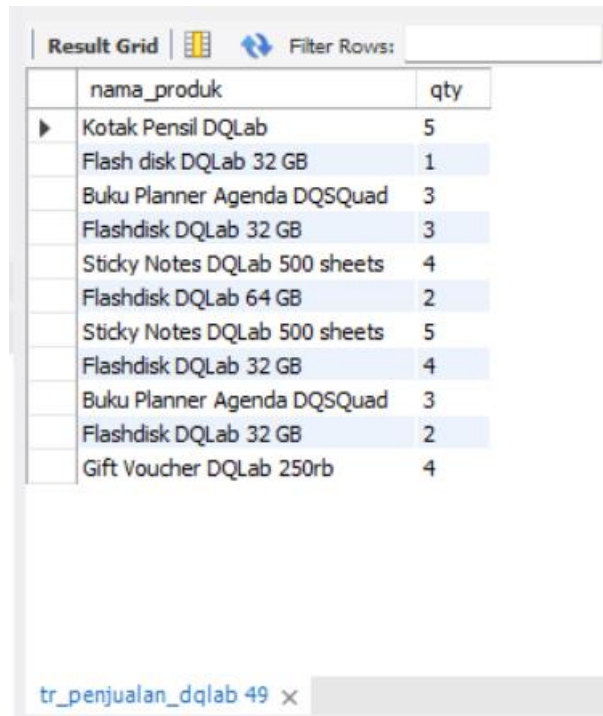
Praktek 16

Filtering dengan operator logika OR

Query:

```
USE dqlab_sales;  
SELECT nama_produk, qty FROM tr_penjualan_dqlab WHERE  
nama_produk LIKE 'f%' OR qty > 2;
```

Output:



The screenshot shows a database query result grid with two columns: 'nama_produk' and 'qty'. The grid contains 13 rows of data. The first row is 'Kotak Pensil DQLab' with a quantity of 5. The second row is 'Flash disk DQLab 32 GB' with a quantity of 1. The third row is 'Buku Planner Agenda DQSquad' with a quantity of 3. The fourth row is 'Flashdisk DQLab 32 GB' with a quantity of 3. The fifth row is 'Sticky Notes DQLab 500 sheets' with a quantity of 4. The sixth row is 'Flashdisk DQLab 64 GB' with a quantity of 2. The seventh row is 'Sticky Notes DQLab 500 sheets' with a quantity of 5. The eighth row is 'Flashdisk DQLab 32 GB' with a quantity of 4. The ninth row is 'Buku Planner Agenda DQSquad' with a quantity of 3. The tenth row is 'Flashdisk DQLab 32 GB' with a quantity of 2. The eleventh row is 'Gift Voucher DQLab 250rb' with a quantity of 4. The twelfth and thirteenth rows are empty. The grid is titled 'Result Grid' and has a 'Filter Rows' button. The table name 'tr_penjualan_dqlab' is visible at the bottom left of the grid.

nama_produk	qty
Kotak Pensil DQLab	5
Flash disk DQLab 32 GB	1
Buku Planner Agenda DQSquad	3
Flashdisk DQLab 32 GB	3
Sticky Notes DQLab 500 sheets	4
Flashdisk DQLab 64 GB	2
Sticky Notes DQLab 500 sheets	5
Flashdisk DQLab 32 GB	4
Buku Planner Agenda DQSquad	3
Flashdisk DQLab 32 GB	2
Gift Voucher DQLab 250rb	4

MODUL 4

ORDER BY, FUNGSI AGREGASI, GROUP BY, HAVING

Penjelasan laporan:

Dokumentasikan seluruh aktivitas praktikum untuk modul 2 dengan menuliskan query dan menempelkan screenshot output dari query tersebut.

Dengan ketentuan:

1. Gunakan font `courier new` untuk menuliskan query
2. Tempelkan Screenshot hasil query dalam kolom output

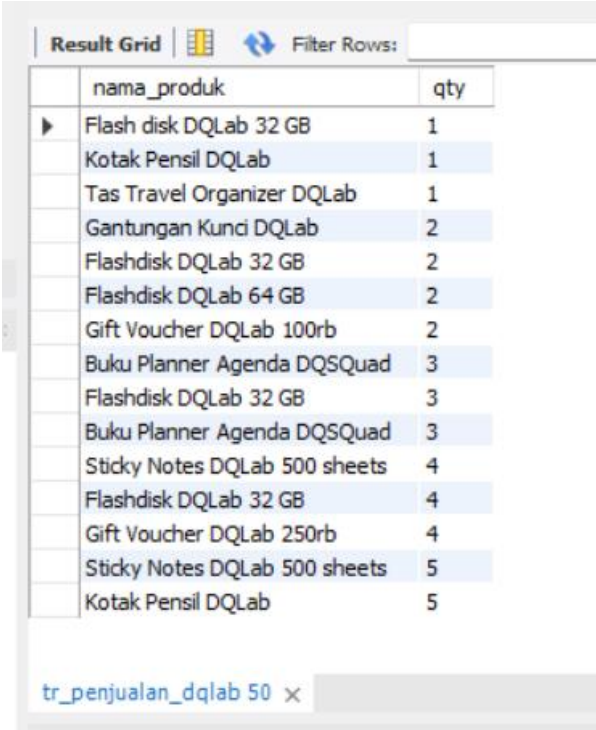
Praktek 1

ORDER BY dengan satu kolom

Query:

```
USE dqlab_sales;  
SELECT nama_produk, qty FROM tr_penjualan_dqlab ORDER  
BY qty;
```

Output:



The screenshot shows a database interface with a 'Result Grid' tab. It displays the output of a SQL query. The grid has two columns: 'nama_produk' and 'qty'. The data is sorted by 'qty' in ascending order. Below the grid, a tab for 'tr_penjualan_dqlab 50' is visible.

nama_produk	qty
Flash disk DQLab 32 GB	1
Kotak Pensil DQLab	1
Tas Travel Organizer DQLab	1
Gantungan Kunci DQLab	2
Flashdisk DQLab 32 GB	2
Flashdisk DQLab 64 GB	2
Gift Voucher DQLab 100rb	2
Buku Planner Agenda DQSQuad	3
Flashdisk DQLab 32 GB	3
Buku Planner Agenda DQSQuad	3
Sticky Notes DQLab 500 sheets	4
Flashdisk DQLab 32 GB	4
Gift Voucher DQLab 250rb	4
Sticky Notes DQLab 500 sheets	5
Kotak Pensil DQLab	5

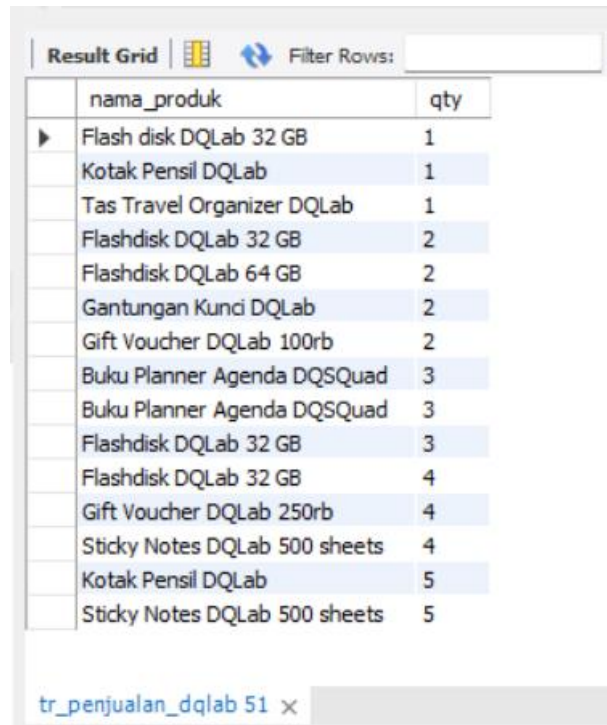
Praktek 2

ORDER BY dengan dua kolom

Query:

```
USE dqlab_sales;  
SELECT nama_produk, qty FROM tr_penjualan_dqlab ORDER  
BY qty, nama_produk;
```

Output:



The screenshot shows a 'Result Grid' window with a table of products and their quantities. The table has two columns: 'nama_produk' and 'qty'. The data is sorted by 'qty' in ascending order, and for products with the same quantity, they are sorted by 'nama_produk' in ascending order. The table contains 18 rows of data.

nama_produk	qty
Flash disk DQLab 32 GB	1
Kotak Pensil DQLab	1
Tas Travel Organizer DQLab	1
Flashdisk DQLab 32 GB	2
Flashdisk DQLab 64 GB	2
Gantungan Kunci DQLab	2
Gift Voucher DQLab 100rb	2
Buku Planner Agenda DQSquad	3
Buku Planner Agenda DQSquad	3
Flashdisk DQLab 32 GB	3
Flashdisk DQLab 32 GB	4
Gift Voucher DQLab 250rb	4
Sticky Notes DQLab 500 sheets	4
Kotak Pensil DQLab	5
Sticky Notes DQLab 500 sheets	5

tr_penjualan_dqlab 51 x

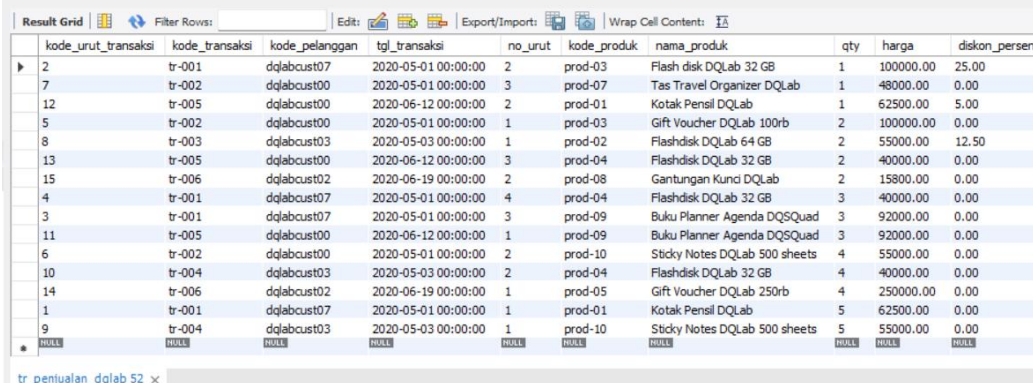
Latihan Mandiri

Soal 1

Query:

```
USE dqlab_sales;  
SELECT * FROM tr_penjualan_dqlab ORDER BY qty,  
tgl_transaksi;
```

Output:



The screenshot shows a database query result grid with the following columns: kode_urut_transaksi, kode_transaksi, kode_pelanggan, tgl_transaksi, no_urut, kode_produk, nama_produk, qty, harga, and diskon_persen. The data is sorted by kode_transaksi. The last row is highlighted with a star icon and labeled 'tr_penjualan_dqlab 52 x'.

kode_urut_transaksi	kode_transaksi	kode_pelanggan	tgl_transaksi	no_urut	kode_produk	nama_produk	qty	harga	diskon_persen
2	tr-001	dqlabcust07	2020-05-01 00:00:00	2	prod-03	Flash disk DQLab 32 GB	1	100000.00	25.00
7	tr-002	dqlabcust00	2020-05-01 00:00:00	3	prod-07	Tas Travel Organizer DQLab	1	48000.00	0.00
12	tr-005	dqlabcust00	2020-06-12 00:00:00	2	prod-01	Kotak Pensil DQLab	1	62500.00	5.00
5	tr-002	dqlabcust00	2020-05-01 00:00:00	1	prod-03	Gift Voucher DQLab 100rb	2	100000.00	0.00
8	tr-003	dqlabcust03	2020-05-03 00:00:00	1	prod-02	Flashdisk DQLab 64 GB	2	55000.00	12.50
13	tr-005	dqlabcust00	2020-06-12 00:00:00	3	prod-04	Flashdisk DQLab 32 GB	2	40000.00	0.00
15	tr-006	dqlabcust02	2020-06-19 00:00:00	2	prod-08	Gantungan Kunci DQLab	2	15800.00	0.00
4	tr-001	dqlabcust07	2020-05-01 00:00:00	4	prod-04	Flashdisk DQLab 32 GB	3	40000.00	0.00
3	tr-001	dqlabcust07	2020-05-01 00:00:00	3	prod-09	Buku Planner Agenda DQSQuad	3	92000.00	0.00
11	tr-005	dqlabcust00	2020-06-12 00:00:00	1	prod-09	Buku Planner Agenda DQSQuad	3	92000.00	0.00
6	tr-002	dqlabcust00	2020-05-01 00:00:00	2	prod-10	Sticky Notes DQLab 500 sheets	4	55000.00	0.00
10	tr-004	dqlabcust03	2020-05-03 00:00:00	2	prod-04	Flashdisk DQLab 32 GB	4	40000.00	0.00
14	tr-006	dqlabcust02	2020-06-19 00:00:00	1	prod-05	Gift Voucher DQLab 250rb	4	250000.00	0.00
1	tr-001	dqlabcust07	2020-05-01 00:00:00	1	prod-01	Kotak Pensil DQLab	5	62500.00	0.00
9	tr-004	dqlabcust03	2020-05-03 00:00:00	1	prod-10	Sticky Notes DQLab 500 sheets	5	55000.00	0.00
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Soal 2

Query:

```
USE kelas_c_mart;  
SELECT * FROM pelanggan ORDER BY nama_pelanggan;
```

Output:



The screenshot shows a database query result grid with the following columns: kode_pelanggan, no_urut, nama_pelanggan, and alamat. The data is sorted by nama_pelanggan. The last row is highlighted with a star icon.

kode_pelanggan	no_urut	nama_pelanggan	alamat
dqlabcust07	7	Agus Cahyono	Vila Gunung Seribu, Blok F4 - No. 8
dqlabcust10	10	Djoko Wardoyo, Drs.	Vila Bukit Sagitarius, Blok A1 No. 1
dqlabcust01	1	Eva Novianti, S.H.	Vila Sempilan, No. 67 - Kota B
dqlabcust02	2	Heidi Goh	Vila Sempilan, No. 11 - Kota B
dqlabcust09	9	Ir. Ita Nugraha	Vila Bukit Sagitarius, Gang Kelapa No. 6
dqlabcust06	6	Irwan Setianto	Vila Gunung Seribu, Blok O1 - No. 1 - Kota C
dqlabcust04	4	Jokolono Sukarman	Vila Permata Intan Berkilau, Blok C5-7
dqlabcust08	8	Maria Sirait	Vila Bukit Sagitarius, Gang. Sawit No. 3
dqlabcust00	0	Pelanggan Non Member	NULL
dqlabcust05	5	Tommy Sinaga	Vila Permata Intan Berkilau, Blok A1/2
dqlabcust03	3	Unang Handoko	Vila Sempilan, No. 1 - Kota B
*	NULL	NULL	NULL

Soal 3

Query:

```
USE kelas_c_mart;  
SELECT * FROM pelanggan ORDER BY alamat;
```

Output:

kode_pelanggan	no_urut	nama_pelanggan	alamat
dqlabcust00	0	Pelanggan Non Member	NULL
dqlabcust10	10	Djoko Wardoyo, Drs.	Vila Bukit Sagitarius, Blok A1 No. 1
dqlabcust09	9	Ir. Ita Nugraha	Vila Bukit Sagitarius, Gang Kelapa No. 6
dqlabcust08	8	Maria Sirait	Vila Bukit Sagitarius, Gang. Sawit No. 3
dqlabcust07	7	Agus Cahyono	Vila Gunung Seribu, Blok F4 - No. 8
dqlabcust06	6	Irwan Setianto	Vila Gunung Seribu, Blok O1 - No. 1 - Kota C
dqlabcust05	5	Tommy Sinaga	Vila Permata Intan Berkilau, Blok A1/2
dqlabcust04	4	Jokolono Sukarman	Vila Permata Intan Berkilau, Blok C5-7
dqlabcust03	3	Unang Handoko	Vila Sempilan, No. 1 - Kota B
dqlabcust02	2	Heidi Goh	Vila Sempilan, No. 11 - Kota B
dqlabcust01	1	Eva Novianti, S.H.	Vila Sempilan, No. 67 - Kota B
NULL	NULL	NULL	NULL

Praktek 3

Menggunakan DESC pada ORDER BY

Query:

```
USE dqlab_sales;  
SELECT nama_produk, qty FROM tr_penjualan_dqlab ORDER  
BY qty DESC;
```

Output:

nama_produk	qty
Kotak Pensil DQLab	5
Sticky Notes DQLab 500 sheets	5
Gift Voucher DQLab 250rb	4
Sticky Notes DQLab 500 sheets	4
Flashdisk DQLab 32 GB	4
Buku Planner Agenda DQSquad	3
Flashdisk DQLab 32 GB	3
Buku Planner Agenda DQSquad	3
Flashdisk DQLab 32 GB	2
Flashdisk DQLab 64 GB	2
Gift Voucher DQLab 100rb	2
Gantungan Kunci DQLab	2
Tas Travel Organizer DQLab	1
Kotak Pensil DQLab	1
Flash disk DQLab 32 GB	1

tr_penjualan_dqlab 55 x

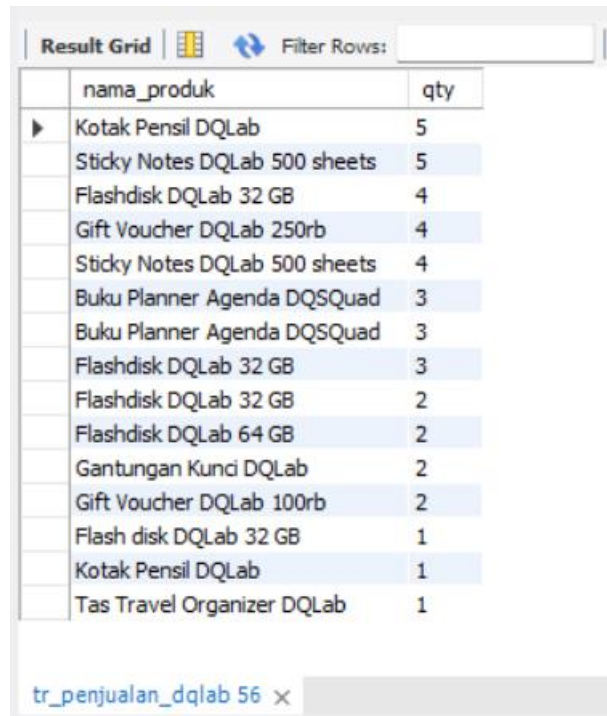
Praktek 4

Menggunakan ASC dan DESC pada dua kolom

Query:

```
USE dqlab_sales;  
SELECT nama_produk, qty FROM tr_penjualan_dqlab ORDER  
BY qty DESC, nama_produk ASC;
```

Output:



	nama_produk	qty
▶	Kotak Pensil DQLab	5
	Sticky Notes DQLab 500 sheets	5
	Flashdisk DQLab 32 GB	4
	Gift Voucher DQLab 250rb	4
	Sticky Notes DQLab 500 sheets	4
	Buku Planner Agenda DQSQuad	3
	Buku Planner Agenda DQSQuad	3
	Flashdisk DQLab 32 GB	3
	Flashdisk DQLab 32 GB	2
	Flashdisk DQLab 64 GB	2
	Gantungan Kunci DQLab	2
	Gift Voucher DQLab 100rb	2
	Flash disk DQLab 32 GB	1
	Kotak Pensil DQLab	1
	Tas Travel Organizer DQLab	1

tr_penjualan_dqlab 56 ×

Latihan Mandiri

Soal 1

Query:

```
USE dqlab_sales;  
SELECT * FROM tr_penjualan_dqlab ORDER BY  
tgl_transaksi DESC, qty ASC;
```

Output:

Result Grid										
Filter Rows:										
Edit: Export/Import: Wrap Cell Content:										
	kode_urut_transaksi	kode_transaksi	kode_pelanggan	tgl_transaksi	no_urut	kode_produk	nama_produk	qty	harga	diskon_persen
15		tr-006	dqlabcust02	2020-06-19 00:00:00	2	prod-08	Gantungan Kunci DQLab	2	15800.00	0.00
14		tr-006	dqlabcust02	2020-06-19 00:00:00	1	prod-05	Gift Voucher DQLab 250rb	4	250000.00	0.00
12		tr-005	dqlabcust00	2020-06-12 00:00:00	2	prod-01	Kotak Pensil DQLab	1	62500.00	5.00
13		tr-005	dqlabcust00	2020-06-12 00:00:00	3	prod-04	Flashdisk DQLab 32 GB	2	40000.00	0.00
11		tr-005	dqlabcust00	2020-06-12 00:00:00	1	prod-09	Buku Planner Agenda DQSQuad	3	92000.00	0.00
8		tr-003	dqlabcust03	2020-05-03 00:00:00	1	prod-02	Flashdisk DQLab 64 GB	2	55000.00	12.50
10		tr-004	dqlabcust03	2020-05-03 00:00:00	2	prod-04	Flashdisk DQLab 32 GB	4	40000.00	0.00
9		tr-004	dqlabcust03	2020-05-03 00:00:00	1	prod-10	Sticky Notes DQLab 500 sheets	5	55000.00	0.00
7		tr-002	dqlabcust00	2020-05-01 00:00:00	3	prod-07	Tas Travel Organizer DQLab	1	48000.00	0.00
2		tr-001	dqlabcust07	2020-05-01 00:00:00	2	prod-03	Flash disk DQLab 32 GB	1	100000.00	25.00
5		tr-002	dqlabcust00	2020-05-01 00:00:00	1	prod-03	Gift Voucher DQLab 100rb	2	100000.00	0.00
4		tr-001	dqlabcust07	2020-05-01 00:00:00	4	prod-04	Flashdisk DQLab 32 GB	3	40000.00	0.00
3		tr-001	dqlabcust07	2020-05-01 00:00:00	3	prod-09	Buku Planner Agenda DQSQuad	3	92000.00	0.00
6		tr-002	dqlabcust00	2020-05-01 00:00:00	2	prod-10	Sticky Notes DQLab 500 sheets	4	55000.00	0.00
1		tr-001	dqlabcust07	2020-05-01 00:00:00	1	prod-01	Kotak Pensil DQLab	5	62500.00	0.00
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

tr_penjualan_dqlab 57 x

Soal 2

Query:

```
USE kelas_c_mart;  
SELECT * FROM pelanggan ORDER BY nama_pelanggan DESC;
```

Output:

Result Grid				
Filter Rows:				
Edit: Export/Import:				
	kode_pelanggan	no_urut	nama_pelanggan	alamat
▶	dqlabcust03	3	Unang Handoko	Vila Sempilan, No. 1 - Kota B
	dqlabcust05	5	Tommy Sinaga	Vila Permata Intan Berkilau, Blok A1/2
	dqlabcust00	0	Pelanggan Non Member	NULL
	dqlabcust08	8	Maria Sirait	Vila Bukit Sagitarius, Gang. Sawit No. 3
	dqlabcust04	4	Jokolono Sukarman	Vila Permata Intan Berkilau, Blok C5-7
	dqlabcust06	6	Irwan Setianto	Vila Gunung Seribu, Blok O1 - No. 1 - Kota C
	dqlabcust09	9	Ir. Ita Nugraha	Vila Bukit Sagitarius, Gang Kelapa No. 6
	dqlabcust02	2	Heidi Goh	Vila Sempilan, No. 11 - Kota B
	dqlabcust01	1	Eva Novianti, S.H.	Vila Sempilan, No. 67 - Kota B
	dqlabcust10	10	Djoko Wardoyo, Drs.	Vila Bukit Sagitarius, Blok A1 No. 1
	dqlabcust07	7	Agus Cahyono	Vila Gunung Seribu, Blok F4 - No. 8
*	NULL	NULL	NULL	NULL

pelanggan 58 x

Soal 3

Query:

```
USE kelas_c_mart;  
SELECT * FROM pelanggan ORDER BY alamat DESC;
```

Output:



	kode_pelanggan	no_urut	nama_pelanggan	alamat
▶	dqlabcust01	1	Eva Novianti, S.H.	Vila Sempilan, No. 67 - Kota B
	dqlabcust02	2	Heidi Goh	Vila Sempilan, No. 11 - Kota B
	dqlabcust03	3	Unang Handoko	Vila Sempilan, No. 1 - Kota B
	dqlabcust04	4	Jokolono Sukarman	Vila Permata Intan Berkilau, Blok C5-7
	dqlabcust05	5	Tommy Sinaga	Vila Permata Intan Berkilau, Blok A1/2
	dqlabcust06	6	Irwan Setianto	Vila Gunung Seribu, Blok O1 - No. 1 - Kota C
	dqlabcust07	7	Agus Cahyono	Vila Gunung Seribu, Blok F4 - No. 8
	dqlabcust08	8	Maria Sirait	Vila Bukit Sagitarius, Gang. Sawit No. 3
	dqlabcust09	9	Ir. Ita Nugraha	Vila Bukit Sagitarius, Gang Kelapa No. 6
	dqlabcust10	10	Djoko Wardoyo, Drs.	Vila Bukit Sagitarius, Blok A1 No. 1
	dqlabcust00	0	Pelanggan Non Mem...	NULL
•	NULL	NULL	NULL	NULL

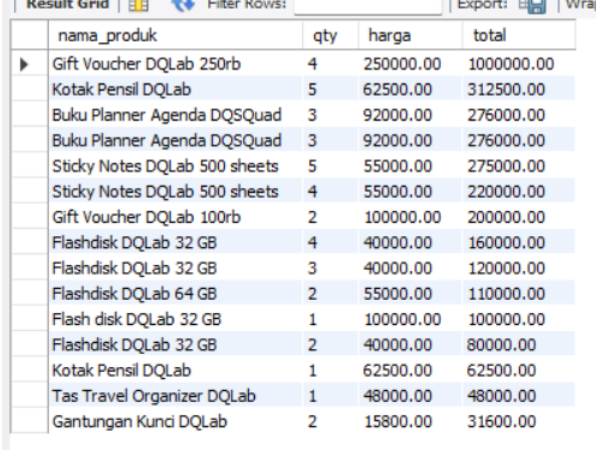
Praktek 5

Menggunakan perhitungan pada ORDER BY

Query:

```
USE dqlab_sales;  
SELECT nama_produk, qty, harga, qty*harga AS total  
FROM tr_penjualan_dqlab ORDER BY harga*qty DESC;
```

Output:



	nama_produk	qty	harga	total
▶	Gift Voucher DQLab 250rb	4	250000.00	1000000.00
	Kotak Pensil DQLab	5	62500.00	312500.00
	Buku Planner Agenda DQSquad	3	92000.00	276000.00
	Buku Planner Agenda DQSquad	3	92000.00	276000.00
	Sticky Notes DQLab 500 sheets	5	55000.00	275000.00
	Sticky Notes DQLab 500 sheets	4	55000.00	220000.00
	Gift Voucher DQLab 100rb	2	100000.00	200000.00
	Flashdisk DQLab 32 GB	4	40000.00	160000.00
	Flashdisk DQLab 32 GB	3	40000.00	120000.00
	Flashdisk DQLab 64 GB	2	55000.00	110000.00
	Flash disk DQLab 32 GB	1	100000.00	100000.00
	Flashdisk DQLab 32 GB	2	40000.00	80000.00
	Kotak Pensil DQLab	1	62500.00	62500.00
	Tas Travel Organizer DQLab	1	48000.00	48000.00
	Gantungan Kunci DQLab	2	15800.00	31600.00

Result 60 ×

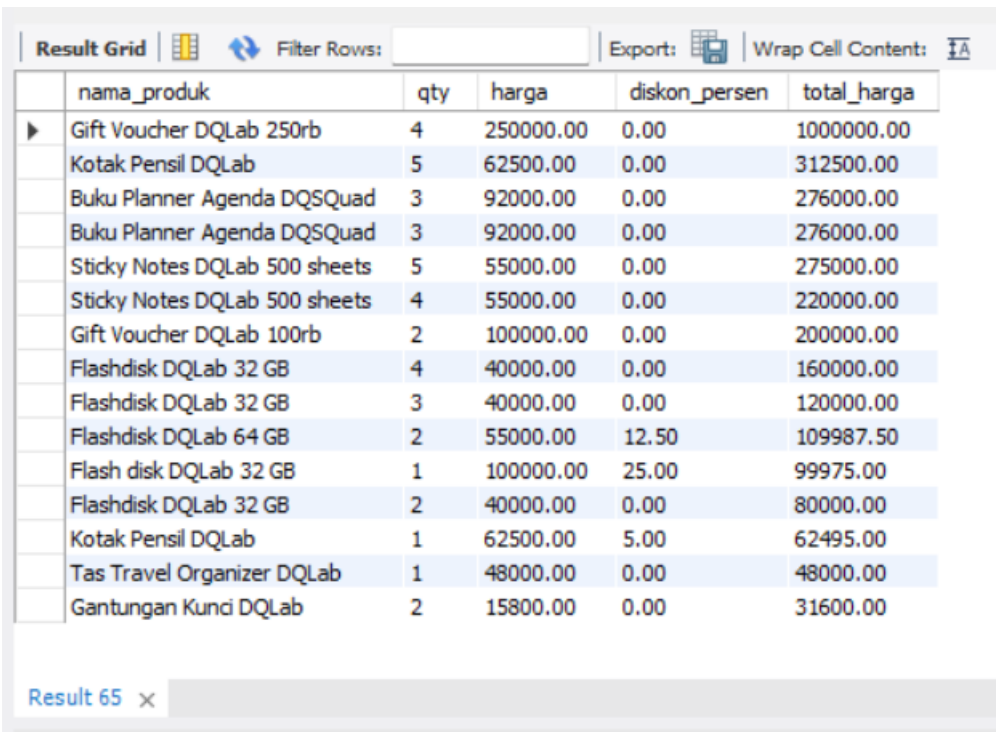
Latihan Mandiri

Cobalah pengurutan dengan ekspresi total harga menggunakan rumusan jumlah barang dikali harga barang dikurangi diskon.

Query:

```
USE dqlab_sales;
SELECT nama_produk, qty, harga, diskon_persen,
(qty*harga)-diskon_persen AS total_harga
FROM tr_penjualan_dqlab ORDER BY (harga*qty)-
diskon_persen DESC;
```

Output:



The screenshot shows a database query result grid with the following columns: nama_produk, qty, harga, diskon_persen, and total_harga. The results are ordered by total_harga in descending order. The grid includes 18 rows of data, with the first row being 'Gift Voucher DQLab 250rb' and the last row being 'Gantungan Kunci DQLab'. The total_harga values range from 1000000.00 down to 31600.00.

	nama_produk	qty	harga	diskon_persen	total_harga
▶	Gift Voucher DQLab 250rb	4	250000.00	0.00	1000000.00
	Kotak Pensil DQLab	5	62500.00	0.00	312500.00
	Buku Planner Agenda DQSQuad	3	92000.00	0.00	276000.00
	Buku Planner Agenda DQSQuad	3	92000.00	0.00	276000.00
	Sticky Notes DQLab 500 sheets	5	55000.00	0.00	275000.00
	Sticky Notes DQLab 500 sheets	4	55000.00	0.00	220000.00
	Gift Voucher DQLab 100rb	2	100000.00	0.00	200000.00
	Flashdisk DQLab 32 GB	4	40000.00	0.00	160000.00
	Flashdisk DQLab 32 GB	3	40000.00	0.00	120000.00
	Flashdisk DQLab 64 GB	2	55000.00	12.50	109987.50
	Flash disk DQLab 32 GB	1	100000.00	25.00	99975.00
	Flashdisk DQLab 32 GB	2	40000.00	0.00	80000.00
	Kotak Pensil DQLab	1	62500.00	5.00	62495.00
	Tas Travel Organizer DQLab	1	48000.00	0.00	48000.00
	Gantungan Kunci DQLab	2	15800.00	0.00	31600.00

Result 65 x

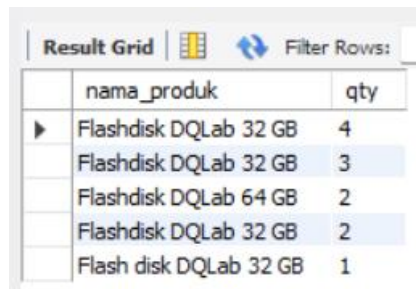
Praktek 6

Menggunakan WHERE pada ORDER BY

Query:

```
USE dqlab_sales;
SELECT nama_produk, qty FROM tr_penjualan_dqlab WHERE
nama_produk LIKE 'F%' ORDER BY qty DESC;
```


Output:



	nama_produk	qty
▶	Flashdisk DQLab 32 GB	4
	Flashdisk DQLab 32 GB	3
	Flashdisk DQLab 64 GB	2
	Flashdisk DQLab 32 GB	2
	Flash disk DQLab 32 GB	1

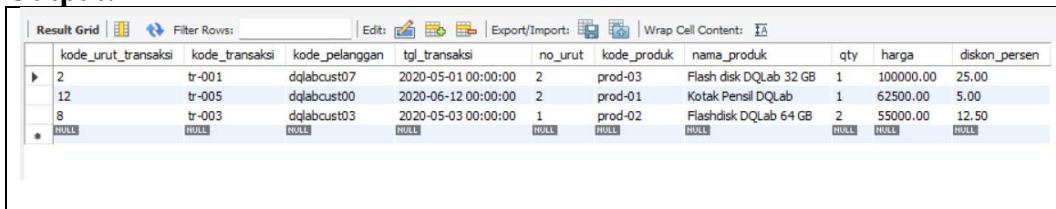
Latihan Mandiri

Soal 1

Query:

```
USE dqlab_sales;  
SELECT * FROM tr_penjualan_dqlab WHERE diskon_persen  
ORDER BY harga DESC;
```

Output:



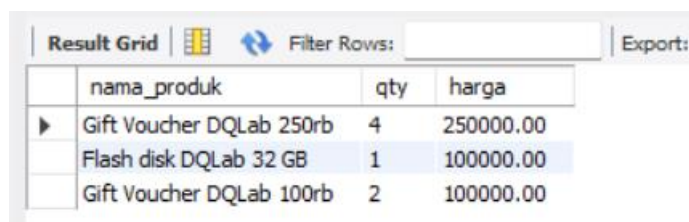
	kode_urut_transaksi	kode_transaksi	kode_pelanggan	tgl_transaksi	no_urut	kode_produk	nama_produk	qty	harga	diskon_persen
▶	2	tr-001	dqlabcust07	2020-05-01 00:00:00	2	prod-03	Flash disk DQLab 32 GB	1	100000.00	25.00
	12	tr-005	dqlabcust00	2020-06-12 00:00:00	2	prod-01	Kotak Pensil DQLab	1	62500.00	5.00
	8	tr-003	dqlabcust03	2020-05-03 00:00:00	1	prod-02	Flashdisk DQLab 64 GB	2	55000.00	12.50
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Soal 2

Query:

```
USE dqlab_sales;  
SELECT nama_produk, qty, harga FROM  
tr_penjualan_dqlab WHERE harga >= 100000 ORDER BY harga  
DESC;
```

Output:



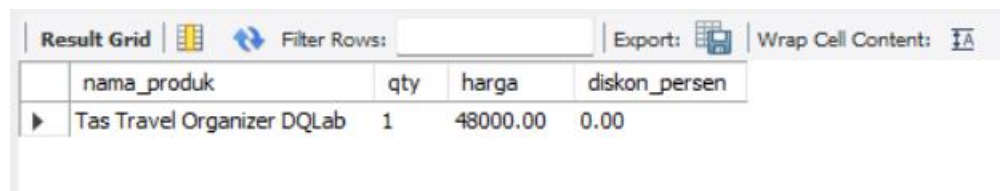
	nama_produk	qty	harga
▶	Gift Voucher DQLab 250rb	4	250000.00
	Flash disk DQLab 32 GB	1	100000.00
	Gift Voucher DQLab 100rb	2	100000.00

Soal 3

Query:

```
USE dqlab_sales;
-- nama produk berawalan karakter 'T' dan diurutkan
berdasarkan diskon tertinggi (jika lebih dari satu
data)
SELECT nama_produk, qty, harga, diskon_persen FROM
tr_penjualan_dqlab WHERE nama_produk LIKE 'T%' ORDER
BY diskon_persen DESC;
```

Output:



The screenshot shows a SQL query result grid with the following data:

	nama_produk	qty	harga	diskon_persen
▶	Tas Travel Organizer DQLab	1	48000.00	0.00

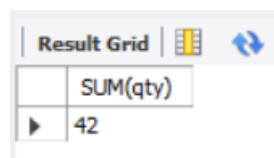
Praktek 7

Menggunakan Fungsi Agregasi SUM

Query:

```
USE dqlab_sales;
SELECT SUM(qty) FROM tr_penjualan_dqlab;
```

Output:



The screenshot shows a SQL query result grid with the following data:

	SUM(qty)
▶	42

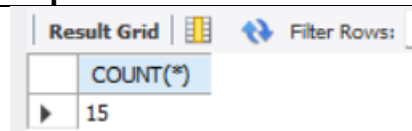
Praktek 8

Menggunakan Fungsi Agregasi COUNT

Query:

```
USE dqlab_sales;
SELECT COUNT(*) FROM tr_penjualan_dqlab;
```

Output:



The screenshot shows a SQL query result grid with the following data:

	COUNT(*)
▶	15

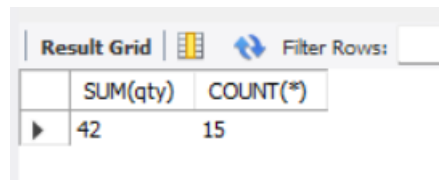
Praktek 9

Menggunakan SUM dan COUNT berbarengan

Query:

```
USE dqlab_sales;  
SELECT SUM(qty), COUNT(*) FROM tr_penjualan_dqlab;
```

Output:



	SUM(qty)	COUNT(*)
▶	42	15

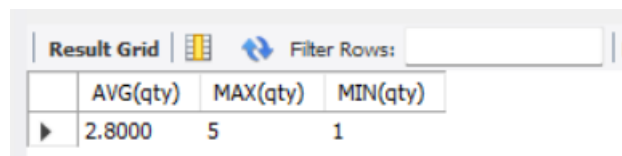
Praktek 10

Menggunakan Fungsi AVG, MAX, dan MIN

Query:

```
USE dqlab_sales;  
SELECT AVG(qty), MAX(qty), MIN(qty) FROM  
tr_penjualan_dqlab;
```

Output:



	AVG(qty)	MAX(qty)	MIN(qty)
▶	2.8000	5	1

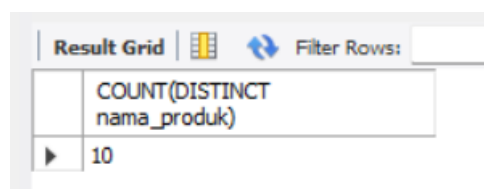
Praktek 11

Menggunakan COUNT(DISTINCT)

Query:

```
USE dqlab_sales;  
SELECT COUNT(DISTINCT nama_produk) FROM  
tr_penjualan_dqlab;
```

Output:



	COUNT(DISTINCT nama_produk)
▶	10

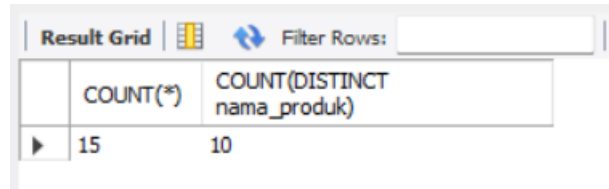
Praktek 12

Menggunakan COUNT(*) dan COUNT(DISTINCT)

Query:

```
USE dqlab_sales;  
SELECT COUNT(*), COUNT(DISTINCT nama_produk) FROM  
tr_penjualan_dqlab;
```

Output:



	COUNT(*)	COUNT(DISTINCT nama_produk)
▶	15	10

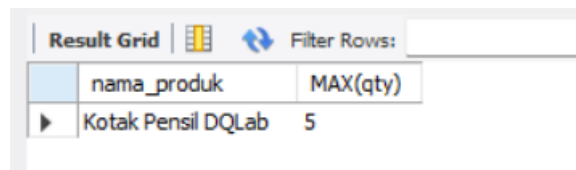
Praktek 13

Penggunaan Field Lain + Fungsi Agregasi

Query:

```
USE dqlab_sales;  
SELECT nama_produk, MAX(qty) FROM tr_penjualan_dqlab;
```

Output:



	nama_produk	MAX(qty)
▶	Kotak Pensil DQLab	5

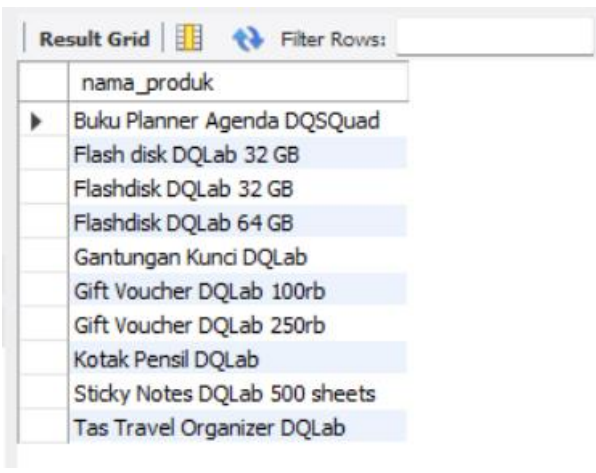
Praktek 14

Penggunaan GROUP BY satu kolom

Query:

```
USE dqlab_sales;  
SELECT nama_produk FROM tr_penjualan_dqlab GROUP BY  
nama_produk;
```

Output:



The screenshot shows a database interface with a 'Result Grid' tab. A table is displayed with the following data:

nama_produk
Buku Planner Agenda DQSquad
Flash disk DQLab 32 GB
Flashdisk DQLab 32 GB
Flashdisk DQLab 64 GB
Gantungan Kunci DQLab
Gift Voucher DQLab 100rb
Gift Voucher DQLab 250rb
Kotak Pensil DQLab
Sticky Notes DQLab 500 sheets
Tas Travel Organizer DQLab

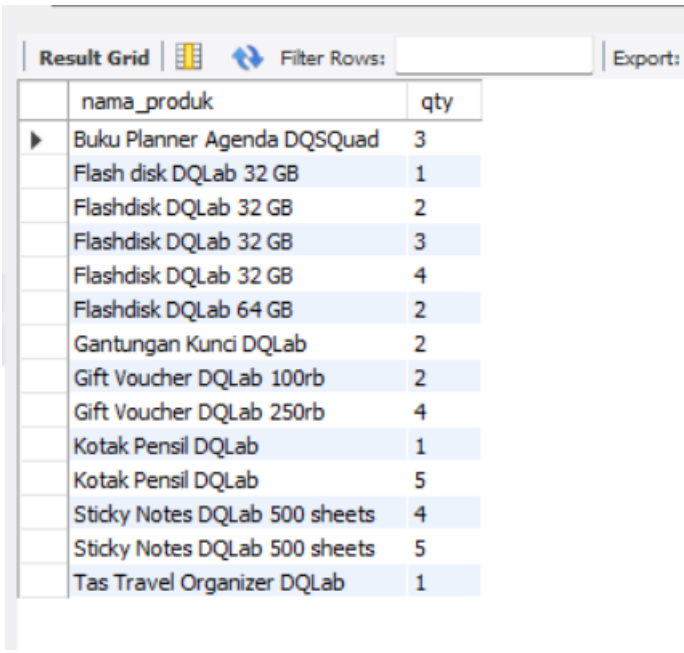
Praktek 15

Penggunaan GROUP BY dua kolom

Query:

```
USE dqlab_sales;  
SELECT nama_produk, qty FROM tr_penjualan_dqlab GROUP  
BY nama_produk, qty;
```

Output:



The screenshot shows a database interface with a 'Result Grid' tab. A table is displayed with the following data:

nama_produk	qty
Buku Planner Agenda DQSquad	3
Flash disk DQLab 32 GB	1
Flashdisk DQLab 32 GB	2
Flashdisk DQLab 32 GB	3
Flashdisk DQLab 32 GB	4
Flashdisk DQLab 64 GB	2
Gantungan Kunci DQLab	2
Gift Voucher DQLab 100rb	2
Gift Voucher DQLab 250rb	4
Kotak Pensil DQLab	1
Kotak Pensil DQLab	5
Sticky Notes DQLab 500 sheets	4
Sticky Notes DQLab 500 sheets	5
Tas Travel Organizer DQLab	1

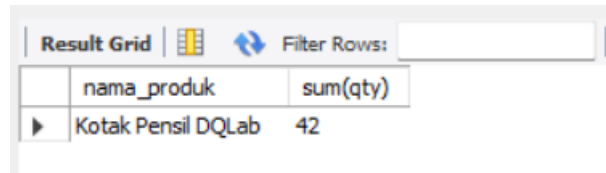
Praktek 16

GROUP BY + Fungsi Agregasi

Query:

```
USE dqlab_sales;  
SELECT nama_produk, sum(qty) FROM tr_penjualan_dqlab;
```

Output:



	nama_produk	sum(qty)
▶	Kotak Pensil DQLab	42

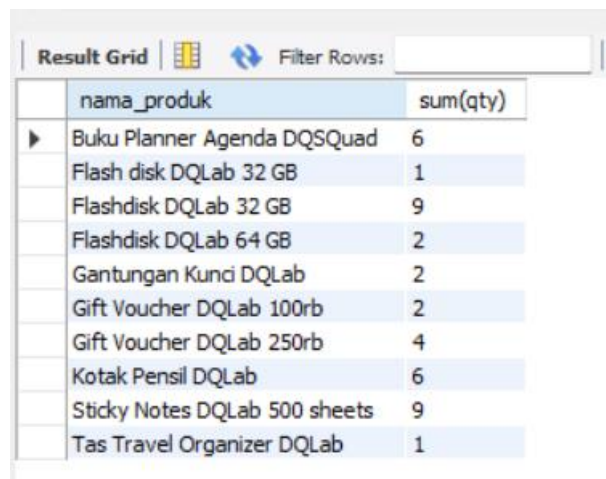
Praktek 17

GROUP BY + Fungsi Agregasi + ORDER BY

Query:

```
USE dqlab_sales;  
SELECT nama_produk, sum(qty) FROM tr_penjualan_dqlab  
GROUP BY nama_produk;
```

Output:



	nama_produk	sum(qty)
▶	Buku Planner Agenda DQSquad	6
	Flash disk DQLab 32 GB	1
	Flashdisk DQLab 32 GB	9
	Flashdisk DQLab 64 GB	2
	Gantungan Kunci DQLab	2
	Gift Voucher DQLab 100rb	2
	Gift Voucher DQLab 250rb	4
	Kotak Pensil DQLab	6
	Sticky Notes DQLab 500 sheets	9
	Tas Travel Organizer DQLab	1

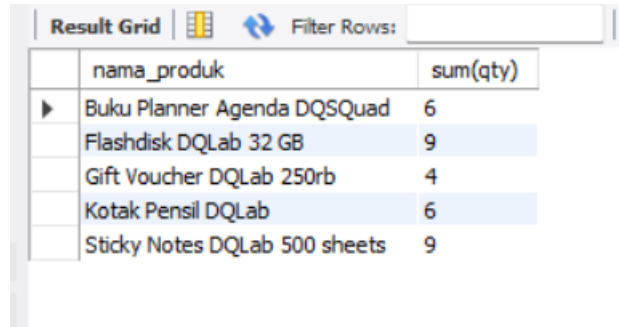
Praktek 18

Penggunaan HAVING

Query:

```
USE dqlab_sales;  
SELECT nama_produk, sum(qty) FROM tr_penjualan_dqlab  
GROUP BY nama_produk HAVING sum(qty) > 2;
```

Output:



	nama_produk	sum(qty)
▶	Buku Planner Agenda DQSQuad	6
	Flashdisk DQLab 32 GB	9
	Gift Voucher DQLab 250rb	4
	Kotak Pensil DQLab	6
	Sticky Notes DQLab 500 sheets	9

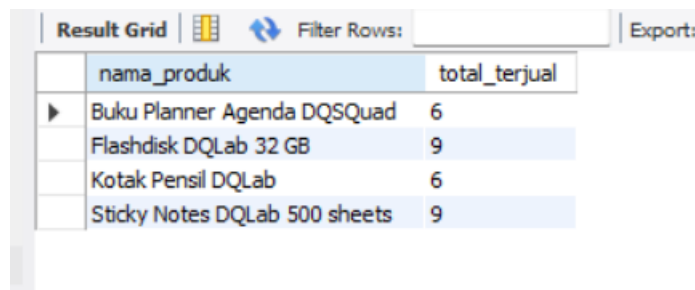
Latihan Mandiri

Soal 1

Query:

```
USE dqlab_sales;  
SELECT nama_produk, sum(qty) AS total_terjual FROM  
tr_penjualan_dqlab GROUP BY nama_produk HAVING  
sum(qty) > 4;
```

Output:



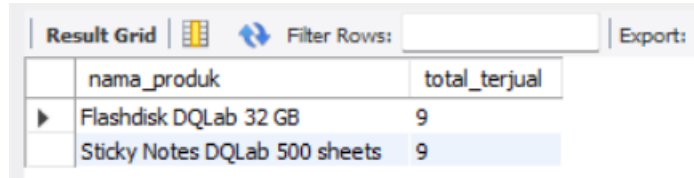
	nama_produk	total_terjual
▶	Buku Planner Agenda DQSQuad	6
	Flashdisk DQLab 32 GB	9
	Kotak Pensil DQLab	6
	Sticky Notes DQLab 500 sheets	9

Soal 2

Query:

```
USE dqlab_sales;  
SELECT nama_produk, sum(qty) AS total_terjual FROM  
tr_penjualan_dqlab GROUP BY nama_produk HAVING  
sum(qty) = 9;
```

Output:



The screenshot shows a database query result grid with two columns: 'nama_produk' and 'total_terjual'. The first row is 'Flashdisk DQLab 32 GB' with a value of 9. The second row is 'Sticky Notes DQLab 500 sheets' with a value of 9. The interface includes a 'Result Grid' tab, a 'Filter Rows' input field, and an 'Export' button.

nama_produk	total_terjual
Flashdisk DQLab 32 GB	9
Sticky Notes DQLab 500 sheets	9

Soal 3

Query:

```
USE dqlab_sales;  
SELECT nama_produk, qty, harga, diskon_persen,  
sum((harga*qty)-diskon_persen)  
AS total_nilai_penjualan FROM tr_penjualan_dqlab  
GROUP BY nama_produk ORDER BY total_nilai_penjualan  
DESC;
```

Output:



The screenshot shows a database query result grid with five columns: 'nama_produk', 'qty', 'harga', 'diskon_persen', and 'total_nilai_penjualan'. The data is sorted in descending order of 'total_nilai_penjualan'. The interface includes a 'Result Grid' tab, a 'Filter Rows' input field, an 'Export' button, and a 'Wrap Cell Content' checkbox.

nama_produk	qty	harga	diskon_persen	total_nilai_penjualan
Gift Voucher DQLab 250rb	4	250000.00	0.00	1000000.00
Buku Planner Agenda DQSQuad	3	92000.00	0.00	552000.00
Sticky Notes DQLab 500 sheets	4	55000.00	0.00	495000.00
Kotak Pensil DQLab	5	62500.00	0.00	374995.00
Flashdisk DQLab 32 GB	3	40000.00	0.00	360000.00
Gift Voucher DQLab 100rb	2	100000.00	0.00	200000.00
Flashdisk DQLab 64 GB	2	55000.00	12.50	109987.50
Flash disk DQLab 32 GB	1	100000.00	25.00	99975.00
Tas Travel Organizer DQLab	1	48000.00	0.00	48000.00
Gantungan Kunci DQLab	2	15800.00	0.00	31600.00